Homological Algebra

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1 Chain Complexes

1.1 Complexes of R-modules

1.1.1 Definition

Let R be a ring. A **chain complex** of R modules is a sequence of R-modules $C_\bullet = \{C_n\}_{n\in\mathbb{Z}}$ along with maps $d_n\colon C_n \to C_{n-1}$ such that $d_{n-1} \circ d_n\colon C_n \to C_{n-2}$ is zero for all n. These maps d_n are called the **differentials** of $C_\bullet$.

We often omit the subscripts for the differentials and the chain complex itself; we write d and C instead of d_n and $C_\bullet$ respectively.

Note that $d_n \circ d_{n+1} = 0$ means that $\mathrm{im} d_{n+1} \subseteq \ker d_n$. This justifies the following definition:

1.1.2 Definition

Given a chain complex C, define

(1) the kernel of $d_n\colon C_n \to C_{n-1}$ to be $Z_n = Z_n(C)$, the n**-cycles** of C;

(2) the image of $d_{n+1}\colon C_{n+1} \to C_n$ to be $B_n = B_n(C)$, the n**-boundaries** of C;

(3) the n**th homology module** of C to be their quotient: Z_n/B_n.

1.1.3 Example

Let $C_n = \mathbb{Z}/8\mathbb{Z}$ for $n \geq 0$ and $C_n = 0$ for $n < 0$. Define d_n to map x to $4x$ (modulo 8).

Then we have $d \circ d$ maps x to $8x \equiv 0$, so this forms a chain complex (of $\mathbb{Z}/8\mathbb{Z}$-modules, say). And Z_n is $\mathbb{Z}/2\mathbb{Z}$, B_n is $\mathbb{Z}/4\mathbb{Z}$. Thus we have H_n is (isomorphic to) $\mathbb{Z}/2\mathbb{Z}$.

We would like to build a category of chain complexes of R-modules. To do so we must define morphisms of chain complexes; there is a clear way to do so.

1.1.4 Definition

A **morphism** of chain complexes $u\colon C \to D$ is a collection of R-module morphisms $u_n\colon C_n \to D_n$ which commute with the differential maps: $u_{n-1} \circ d_n = d_{n-1} \circ u_n$.

The category $\mathsf{Ch}(\mathrm{Mod}_R)$ is the category of chain complexes of (right) R-modules.

1.1.5 Proposition

A morphism of chain complexes $u\colon C \to D$ maps boundaries to boundaries and cycles to cycles. Thus defines maps $H_n(C) \to H_n(D)$. Furthermore, for each $n \in \mathbb{Z}$, H_n defines a functor $\mathsf{Ch}(\mathrm{Mod}_R) \to \mathrm{Mod}_R$.

Proof

We have that $udx = dux$, thus for $dx \in B(C)$ its image under u is in $B(D)$. Similarly if $dx = 0$, then $udx = dux = 0$, in particular $ux \in Z(D)$. And so u_n factors through $B_n(C)$ and restricts to cycles, giving us maps $u_n\colon H_n(C) \to H_n(D)$.

This is a functor, as quotienting a morphism is. ■

1.1.6 Example

Let k be a field, and consider k-vector spaces $\{B_n, H_n\}_{n\in\mathbb{Z}}$. Define $C_n = B_n \oplus H_n \oplus B_{n-1}$ and define $d_n\colon C_n \to C_{n-1}$ by projection onto $B_{n-1} \subseteq C_{n-1}$. This is a chain complex: d_n projects onto B_{n-1} which is orthogonal to B_{n-2}, onto which d_{n-1} projects. Further note that $Z_n(C) = B_n \oplus H_n$ and $B_n(C) = B_n$, thus $H_n(C) = H_n$; justifying the choice of names.

If V is a chain complex of k-vector spaces, then note that $V_n \cong B_n(V) \oplus H_n(V) \oplus B_{n-1}(V)$. Thus all chain complexes of k-vector spaces can be constructed in this manner.

1.1.7 Example

Let C be a chain complex of R-modules, and A an R-module. Then we define $d'_n\colon \hom_R(A, C_n) \to \hom_R(A, C_{n-1})$ by mapping f to $d_n f$. This forms a chain complex clearly.

Now if $A = Z_n(C)$ for some n, suppose $H_n(\hom_R(Z_n, C)) = 0$. This means that $B_n(\hom_R(Z_n, C)) = Z_n(\hom_R(Z_n, C))$, i.e. if $df = 0$ then $f = dg$ for all $f\colon Z_n \to C_n$. In particular for inclusion $\iota\colon Z_n \to C_n$; $d\iota = 0$ so $\iota = df$ for some $f\colon Z_n \to C_{n+1}$. But this means that $Z_n \subseteq B_n$ (and thus there is equality): $Z_n = \mathrm{im}\iota = \mathrm{im}df \subseteq \mathrm{im}d = B_n$, and so $H_n(C) = 0$.

1.1.8 Definition

A morphism of chain complexes $C \to D$ is a **quasi-isomorphism** if the maps $H_n(C) \to H_n(D)$ are all isomorphisms.

There is a clear chain complex for all R: the zero chain complex, consisting of only the zero R-module. This complex is a zero object in $\mathsf{Ch}(\mathrm{Mod}_R)$: it is both initial and final.

1.1.9 Proposition

The following are equivalent for every chain complex C:

(1) C is exact;

(2) C is **acyclic**: $H_n(C) = 0$ for all n;

(3) The map $0 \to C$ is a quasi-isomorphism.

Proof

(1) and (2) are clearly equivalent: $H_n(C) = 0$ iff $B_n(C) = Z_n(C)$, which is just what it means for C to be exact. These are also clearly equivalent to (3), by definition. ■

1.1.10 Definition

A **cochain complex** of R-modules is $C^\bullet$, consisting of a collection of R-modules $\{C^n\}_{n\in\mathbb{Z}}$ along with differential maps $d^n\colon C^n \to C^{n+1}$ such that $d \circ d = 0$. As before, $Z^n(C^\bullet) = \ker d^n$ is the module of **n-cocycles** and $B^n(C^\bullet) = \mathrm{im}d^{n-1} \subseteq C^n$ is the module of **n-coboundaries**. $H^n(C^\bullet) = Z^n(C^\bullet)/B^n(C^\bullet)$ is

the n**th cohomology module** of $C^\bullet$.

Morphisms of cochain complexes are defined the same.

If $C_\bullet$ is a chain complex we can define a cochain complex $C^\bullet$ simply by $C^n = C_{-n}$ and $d^n = d_{-n}$.

1.1.11 Definition

A chain complex C is **bounded** if all but finitely many C_n are zero. If $C_n = 0$ for all n other than $a \leq n \leq b$, then C has **amplitude in** $[a, b]$. C is **bounded above** (resp. **bounded below**) if there is a b (resp. a) such that $C_n = 0$ for $n > b$ (resp. $n < a$).

The full subcategories of bounded, bounded above, bounded below chain complexes are denoted Ch_b, Ch_-, Ch_+ respectively. The subcategory of non-negative chain complexes is denoted by $\mathsf{Ch}_{\geq 0}$ (those chain complexes bounded below by 0).

Similar for cochain complexes.

1.2 Operations on chain complexes

The purpose of this section is to show that chain complexes form an **Abelian category**. Recall the following category-theoretic definitions:

1.2.1 Definition

A **preadditive category** is a category whose hom-sets are given the structure of Abelian groups, such that composition distributes over addition: $h(g+g')f = hgf + hg'f$. An **additive functor** is a functor between two preadditive categories whose restrictions to the hom-sets forms an Abelian group homomorphism.

Finite products and coproducts in a preadditive category necessarily coincide; they are called **biproducts**. An **additive category** is a preadditive category with all finite biproducts (equivalently a zero object and binary biproducts).

$\mathsf{Ch} = \mathsf{Ch}(\mathrm{Mod}_R)$ is an additive category. We can add chain complex morphisms in a clear manner: $(u+v)_n = u_n + v_n$. Furthermore as discussed, Ch has a zero object. Ch has all arbitrary direct products and coproducts (and thus all finite biproducts, so it is an additive category).

Indeed, let $\{C_\alpha\}_{\alpha \in I}$ be a collection of chain complexes of R-modules. We define their product $\prod_\alpha C_\alpha$ to have as components the direct product of the components: the nth component is $\prod_\alpha C_{\alpha,n}$. Similarly their coproduct $\bigoplus_\alpha C_\alpha$'s nth component is $\bigoplus_\alpha C_{\alpha,n}$. The differential maps are defined coordinate-wise as well. These satisfy the universal properties of products and coproducts respectively.

1.2.2 Proposition

Products and coproducts commute with the homology functors. That is to say, $H_n(\prod_\alpha C_\alpha) \cong \prod_\alpha H_n(C_\alpha)$ and similar for coproducts.

1.2.3 Definition

A chain complex B is a **subcomplex** of another chain complex C if B_n is a submodule of C_n for every n, and B's differential maps are the restrictions of C's.

If B is a subcomplex of C, then we can define their **quotient complex** C/B whose components are C_n/B_n and whose differential maps are the factorizations of C and B's.

In particular, if $f\colon B \to C$ is a chain map, the kernels $\{\ker f_n\}_n$ form a subcomplex of B denoted $\ker f$. And the cokernels $\{\operatorname{coker} f_n\}_n$ form a quotient complex of C denoted $\operatorname{coker} f$.

1.2.4 Definition

In a category with a zero object (more generally with zero morphisms), the **kernel** of a morphism f is its equalizer with 0. Its **cokernel** is its coequalizer with 0.

In a general category, a **monomorphism** is a morphism which is left-cancelative: $fg = fh$ implies $g = h$. An **epimorphism** is one which is right-cancelative: $gf = hf$ implies $g = h$. In a preadditive category monomorphisms are precisely those morphisms for which $fg = 0$ implies $g = 0$; epimorphisms are those for which $gf = 0$ implies $g = 0$.

Our notion of kernels and cokernels in Ch coincide with the category-theoretic definitions. Furthermore a monomorphism in Ch is precisely a chain morphism whose components are all monomorphisms, similar for epimorphisms.

1.2.5 Definition

An **Abelian category** is an additive category such that

(1) every map has a kernel and cokernel,

(2) every monomorphism is the kernel of its cokernel,

(3) every epimorphism is the cokernel of its kernel.

The prototypical example of an Abelian category is Mod_R (in particular $\mathrm{Ab} = \mathrm{Mod}_{\mathbb{Z}}$).

1.2.6 Definition

Let $\mathcal{C}$ be a category, and $A \in \mathcal{C}$ an object. We say that two monomorphisms $u\colon X \to A$ and $v\colon Y \to A$ are equivalent if there exists an isomorphism $\phi\colon X \to Y$ such that $u = v\phi$. The equivalence classes of these monomorphisms are called **subobjects** of A.

Similarly two epimorphisms $u\colon A \to X$ and $v\colon A \to Y$ are equivalent if there exists an isomorphism $\phi\colon X \to Y$ such that $v = \phi u$. The equivalence classes of these epimorphisms are called **quotient objects** of A.

1.2.7 Definition

In an Abelian category, the **image** of a morphism $f\colon B \to C$ is $\mathrm{im} f = \ker(\mathrm{coker} f)$, considered as a subobject of C. Its **coimage** is $\mathrm{coim} f = \mathrm{coker}(\ker f)$, viewed as a subobject of B.

In Mod_R we define the image of $f\colon B \to C$ to be $\mathrm{im} f = \{fb \mid b \in B\}$ (the coimage is $B/\ker f$). This coincides with the category-theoretic definition, taking the monomorphism to be the inclusion $\mathrm{im} f \to C$. Every map f of R-modules factors as

$$B \xrightarrow{e} \mathrm{im} f \xrightarrow{m} C$$

with e epi and m monic.

1.2.8 Definition

In an Abelian category, a sequence of morphisms $A \xrightarrow{f} B \xrightarrow{g} C$ is **exact** at B if $\mathrm{im} f = \ker g$ (as subobjects of B).

Note that if $\mathcal{A}$ is an Abelian category, we can repeat the discussion of the previous section to define the category of chain complexes on $\mathcal{A}$, $\mathsf{Ch}(\mathcal{A})$. This is an additive category, and homology are additive functors $\mathsf{Ch}(\mathcal{A}) \to \mathcal{A}$.

1.2.9 Theorem

For an Abelian category $\mathcal{A}$, $\mathsf{Ch}(\mathcal{A})$ is itself Abelian as well.

Proof

$\mathsf{Ch}(A)$ has binary biproducts and a zero object; their constructions are the same as before. Now if $f\colon B \to C$ is a chain map, then f is monic iff $f_n\colon B_n \to C_n$ is monic for all n. Indeed, the composite $\ker f \to C$ (inclusion composed with f) is zero, and so if f is monic then $\ker f = 0$ (since $f\iota = 0$ implies $\iota = 0$), and conversely (recall $\ker f = 0$ iff f is monic). Thus if f is monic, then f is isomorphic to the kernel of $C \to C/B$. Similar for epimorphisms. ■

Since $\mathsf{Ch}(\mathcal{A})$ is itself Abelian, we can form the category of chain complexes over $\mathsf{Ch}(\mathcal{A})$. These can be characterized more elegantly as follows:

1.2.10 Definition

A **bicomplex** in $\mathcal{A}$ is a family $\{C_{p,q}\}_{p,q\in\mathbb{Z}}$ of objects from $\mathcal{A}$ along with maps $d^h\colon C_{p,q} \to C_{p-1,q}$ and $d^v\colon C_{p,q} \to C_{p,q-1}$ (horizontal and vertical differentials) such that $d^h \circ d^h = d^v \circ d^v = d^v \circ d^h + d^h \circ d^v = 0$.

A bicomplex is **bounded** if C has only finitely many nonzero components along every diagonal $p + q = n$ (e.g. if $C_{p,q} \neq 0$ only when $p, q \geq 0$).

If C_{pq} is a bicomplex, $C_{\bullet q}$ and $C_{p\bullet}$ are chain complexes for all p, q.

Note that due to anticommutativity ($d^v d^h = -d^h d^v$), d^v are not maps $C_{\bullet q} \to C_{\bullet q-1}$, but they can be used to construct them: define

$$f_{pq} = (-1)^p d^v_{pq}\colon C_{pq} \to C_{p,q-1}$$

Then $f_{\bullet q}$ defines a chain map $C_{\bullet q} \to C_{\bullet q-1}$: $d^h f_{pq} = d^h(-1)^p d^v_{pq} = (-1)^{p-1} d^v_{p-1,q} d^h = f_{p-1,q} d^h$. This "sign trick" allows us to identify the category of bicomplexes with $\mathsf{Ch}(\mathsf{Ch}(\mathcal{A}))$.

Let C be a bicomplex, and define the chain complexes $\mathrm{Tot}^\Pi(C)$ and $\mathrm{Tot}^\oplus(C)$ by

$$\mathrm{Tot}^\Pi(C)_n = \prod_{p+q=n} C_{pq}, \qquad \mathrm{Tot}^\oplus(C)_n = \bigoplus_{p+q=n} C_{pq}$$

Then $d = d^h + d^v$ defines maps $\mathrm{Tot}^\Pi(C)_n \to \mathrm{Tot}^\Pi(C)_{n-1}$ and $\mathrm{Tot}^\oplus(C)_n \to \mathrm{Tot}^\oplus(C)_{n-1}$. Explicitly, we map $x \in C_{pq}$ to $(d^h x, d^v x) \in C_{p-1,q} \times C_{p,q-1}$, or more suggestively:

$$d \sum_{p+q=n} x_{pq} = \sum_{p+q=n} (d^h x_{pq} + d^v x_{pq})$$

These are differential maps: $(d^h + d^v)(d^h + d^v) = 0$ precisely because $d^h d^v + d^v d^h = 0$.

And so $\mathrm{Tot}^\Pi(C)$ and $\mathrm{Tot}^\oplus(C)$ define chain complexes, called **total complexes**. If C is bounded, then $\mathrm{Tot}^\Pi(C) = \mathrm{Tot}^\oplus(C)$ (as there are only finitely many nonzero C_{pq} for which $p + q = n$).

Note:

The total complexes do not exist in all Abelian categories! In order form them to exist, the infinite products and coproducts of C_{pq} must exist. If C is bounded, then they do, as the total complexes are equal to the finite biproducts of C_{pq}. But in general, we need $\mathcal{A}$ to be complete (equivalently have arbitrary products) to define $\mathrm{Tot}^\Pi(C)$, and cocomplete (arbitrary coproducts) to define $\mathrm{Tot}^\oplus(C)$.

In particular the total complexes exist in Mod_R and $\mathsf{Ch}(\mathrm{Mod}_R)$.

1.2.11 Definition

Let C be a chain complex, and n an integer. We define $\tau_{\geq n}C$ to be the subcomplex of C defined by

$$(\tau_{\geq n}C)_i = \begin{cases} 0 & \text{if } i < n \\ Z_n & \text{if } i = n \\ C_n & \text{if } i > n \end{cases}$$

This is called the **truncation** of C below n. The quotient complex $\tau_{<n}C = C/\tau_{\geq n}C$ is the truncation of C above n.

Note that the ith boundary module and ith cycle module for $\tau_{\geq n}C$ are the same as those for C, except when $i < n$ when they are zero. Thus

$$H_i(\tau_{\geq n}C) = \begin{cases} 0 & i < n \\ H_i(C) & i \geq n \end{cases}$$

And so we get

$$H_i(\tau_{<n}C) = \begin{cases} H_i(C) & i < n \\ 0 & i \geq n \end{cases}$$

The more naive method defining truncation does not give as clean of a homology group (defining the nth term to be 0 or C_n, whatever).

1.2.12 Definition

If C is a chain or cochain complex and p an integer we define a new complex (chain or cochain) $C[p]$ by

$$C[p]_n = C_{n+p} \qquad (\text{resp. } C[p]^n = C^{n-p})$$

with differential $(-1)^p d$. These are called the **pth translate** of C. The sign convention (multiplying by $(-1)^p$ will simplify notation later on).

Clearly we have

$$H_n(C[p]) = H_{n+p}(C) \qquad (\text{resp. } H^n(C[p]) = H^{n-p}(C))$$

We extand translation into a functor by shifting indices on a chain map: if $f\colon C \to D$ is a chain map then we define $f[p]$ by

$$f[p]_n = f_{n+p} \qquad (\text{resp. } f[p]^n = f^{n-p})$$

1.2.13 Example

Let C be a complex, then there is an exact sequence of complexes

$$0 \to Z(C) \to C \xrightarrow{d} B(C)[-1] \to 0$$

Indeed, inclusion is mono and thus has trivial kernel; by definition the kernel of d is the image of $Z(C)$. And the image of d is by definition all boundaries. We must reindex for d to be a chain map.

Similarly we have an exact sequence

$$0 \to H(C) \to C/B(C) \xrightarrow{d} Z(C)[-1] \to H(C)[-1] \to 0$$

1.3 Long exact sequences

Chain complexes are useful as they provide long exact sequences. Our goal in this section is to prove the following:

1.3.1 Theorem

Let $0 \to A \xrightarrow{f} B \xrightarrow{g} C \to 0$ be a short exact sequence of chain complexes. Then there are natural maps $\partial_n \colon H_n(C) \to H_{n-1}(A)$ called **connecting homomorphisms** such that

$$\cdots \xrightarrow{g} H_{n+1}(C) \xrightarrow{\partial} H_n(A) \xrightarrow{f} H_n(B) \xrightarrow{g} H_n(C) \xrightarrow{\partial} H_{n-1}(A) \xrightarrow{f} \cdots$$

is an exact sequence.

Similarly if $0 \to A \xrightarrow{f} B \xrightarrow{g} C \to 0$ is a short exact sequence of cochain complexes, then there are natural maps $\partial_n \colon H^n(C) \to H^{n+1}(A)$ and a long exact sequence:

$$\cdots \xrightarrow{g} H^{n-1}(C) \xrightarrow{\partial} H^n(A) \xrightarrow{f} H^n(B) \xrightarrow{g} H^n(C) \xrightarrow{\partial} H^{n+1}(A) \xrightarrow{f} \cdots$$

1.3.2 Example

Note that if $0 \to A \to B \to C \to 0$ is a short exact sequence such that two of the three complexes A, B, C are exact (acyclic) then so is the third. If A, C are exact then since we have an exact sequence $0 = H_n(A) \to H_n(B) \to H_n(C) = 0$, we must have $H_n(B) = 0$ so B is exact. The other cases are due to the above theorem: we will always have an exact sequence of the form $0 \to H_n(X) \to 0$ for $X = A, B, C$.

We will require the following lemma:

1.3.3 Lemma (Snake lemma)

Consider a commutative diagram of R-modules of the form:

$$\begin{tikzcd}
 & A' \arrow[r, "\alpha"'] \arrow[d, "f"'] & B' \arrow[r, "p"'] \arrow[d, "g"'] & C' \arrow[r] \arrow[d, "h"'] & 0 \\
0 \arrow[r] & A \arrow[r, "i"] & B \arrow[r, "\beta"] & C &
\end{tikzcd}$$

If the rows are exact, then there is an exact sequence

$$0 \to \ker f \to \ker g \to \ker h \xrightarrow{\partial} \operatorname{coker} f \to \operatorname{coker} g \to \operatorname{coker} h \to 0$$

where ∂ is defined by $\partial(c) = i^{-1}gp^{-1}(c)$ for $c \in \ker h$. Moreover, if $\alpha \colon A' \to B'$ is monic then so is $\ker f \to \ker g$; if $\beta \colon B \to C$ is onto then so is $\operatorname{coker} g \to \operatorname{coker} h$.

Proof

The top row is exact, so p is surjective. Thus for $c \in \ker h$ there is an $x \in B'$ such that $p(x) = c$. Note then that $\beta g x = hpx = hc = 0$ and so $gx \in \ker\beta = \operatorname{im} i$ so $i^{-1}gx$ exists. But this doesn't ensure that this is well-defined: if $p(x) = p(y) = c$ we still need to show that $i^{-1}gx = i^{-1}gy$.

Note that i^{-1} here maps into the cokernel of f, that is to say we map to $i^{-1} \bullet + \operatorname{im} f$. So a more exact way of writing ∂ would be by

$$\partial(c) = i^{-1}gp^{-1}(c) + \operatorname{im} f$$

Suppose $px = 0$ then $x \in \ker p = \operatorname{im}\alpha$ so $x = \alpha(a)$ for $a \in A'$. Then we have $gx = ifa$ and so $i^{-1}gx = fa \in \operatorname{im} f$, and so $i^{-1}gx = 0$ in $\operatorname{coker} f$. Thus if $p(x) = p(y)$ then $p(x - y) = 0$ and so $i^{-1}gx - i^{-1}gy \in \operatorname{im} f$ meaning

they define the same coset in $\operatorname{coker} f$; ∂ is well-defined. It is also clearly a homomorphism, as if $p(x) = c$ and $p(y) = c'$ then $\partial(c + c') = i^{-1}gx + i^{-1}gy = \partial(c) + \partial(c')$.

Now $0 \to \ker f \xrightarrow{\alpha} \ker g \xrightarrow{p} \ker h$ is an exact sequence, and so is $\operatorname{coker} f \xrightarrow{i} \operatorname{coker} g \xrightarrow{\beta} \operatorname{coker} h \to 0$. Adding ∂ allows us to connect these two exact sequences. ■

We can extend the snake lemma to arbitrary Abelian categories. The Freyd-Mitchell embedding theorem allows us to embed small Abelian categories into categories of modules. So take the Abelian subcategory generated by the commutative, embed it into a category of modules, and the result is given (details are omitted).

Now we construct the natural maps $\partial\colon H_n(C) \to H_{n-1}(A)$ in our goal theorem, associated with $0 \to A \to B \to C \to 0$. Consider the commutative diagram (we write dA_{n+1} for $B_n(A)$ since we already have a chain complex B_n)

$$\begin{tikzcd}
 & A_n/dA_{n+1} \arrow[r] \arrow[d, "d_A"'] & B_n/dB_{n+1} \arrow[r] \arrow[d, "d_B"'] & C_n/dC_{n+1} \arrow[r] \arrow[d, "d_C"'] & 0 \\
0 \arrow[r] & Z_{n-1}(A) \arrow[r] & Z_{n-1}(B) \arrow[r] & Z_{n-1}(C) &
\end{tikzcd}$$

The rows are exact. The kernel of d_A is $H_n(A)$, its cokernel is $H_{n-1}(A)$. Thus the snake lemma yields an exact sequence

$$H_n(A) \to H_n(B) \to H_n(C) \xrightarrow{\partial} H_{n-1}(A) \to H_{n-1}(B) \to H_{n-1}(C)$$

pasting these sequences together gives us the desired long exact sequence. ■

Using the snake lemma's construction, we can understand ∂. Consider the short exact sequence

$$0 \to A \xrightarrow{f} B \xrightarrow{g} C \to 0$$

then $\partial(z) = f^{-1}dg^{-1}(z)$. That is, for $[z] \in H_n(C)$ consider its lift $b \in B_n$ (meaning $g(b) = z$), apply d and then f^{-1}: $\partial(z) = f^{-1}(db)$.

Now ∂ is natural in the following sense: consider the category $\mathcal{S}$ whose objects are short exact sequences of chain complexes in $\mathcal{A}$, and whose morphisms commute with these. Similarly there is a category $\mathcal{L}$ of long exact sequences in $\mathcal{A}$.

1.3.4 Proposition

The long exact sequence is a functor from $\mathcal{S}$ to $\mathcal{L}$.

We omit the proof here; it is a direct result of the construction of the connecting morphisms (from the snake lemma) and definitions.

1.3.5 Example

Suppose $f\colon C \to D$ is a morphism of chain complexes such that $\ker f, \operatorname{coker} f$ are acylic. Then f is a quasi-isomorphism. Indeed, we have an exact sequence

$$\ker f \to C \xrightarrow{f} D \to \operatorname{coker} f$$

This gives us an exact sequence of homology modules

$$0 = H_n(\ker f) \to H_n(C) \xrightarrow{f} H_n(D) \to H_n(\operatorname{coker} f) = 0$$

meaning that the induced morphism between homology modules must be an isomorphism: f is quasi-isomorphic.

1.3.6 Example

Let $0 \to A \xrightarrow{f} B \xrightarrow{g} C \to 0$ be a short exact sequence of bicomplexes. This gives us a short exact sequence of total complexes. Indeed, f induces a map $\mathrm{Tot}(A) \to \mathrm{Tot}(B)$ by mapping (a_{pq}) to $(f_{pq}a_{pq})$; the induced maps still form into a short exact sequence. Thus if $\mathrm{Tot}(A)$ or $\mathrm{Tot}(C)$ are acylic, then the other two are quasi-isomorphic (since the exact sequence of homology modules is of the form $0 \to X \to Y \to 0$).

1.4 Chain homotopies

Recall that chain complexes of vector spaces over a field decompose as $C_n = Z_n \oplus B'_n$ where $B'_n \cong C_n/Z_n \cong d(C_n) = B_{n-1}$, and $Z_n = B_n \oplus H'_n$ where $H'_n \cong Z_n/B_n = H_n(C)$. So we have compositions

$$C_n \xrightarrow{\text{proj}} Z_n \xrightarrow{\text{proj}} B_n \cong B'_{n+1} \subseteq C_{n+1}$$

giving us splitting maps $s_n\colon C_n \to C_{n+1}$ such that $d = dsd$:

$$C_{n+1} \xrightarrow{d} C_n \to Z_n \to B_n \cong B'_{n+1} \subseteq C_{n+1} \xrightarrow{d} C_n$$

the map $C_{n+1} \to B_n$ simply maps a to da. The isomorphism $B_n \cong B'_{n+1}$ maps da to a's projection onto B'_{n+1} (orthogonal to Z'_{n+1}; following the isomorphisms), which d then maps to da (since da is equal to d on the projection). Thus $dsd = d$, as desired.

The composition ds is the projection of C_n onto B_n, and sd the projection onto B'_n. Thus $ds + sd$ is an endomorphism of C_n whose kernel is H'_n, isomorphic to the homology $H_n(C)$. Its image is $B_n \oplus B'_n$ and as such its cokernel is isomorphic to $H_n(C)$ as well. So the kernel and cokernel of $ds + sd$ are isomorphic to the homology complex $H_\bullet(C)$ (the differentials here are zero, so the homology modules are the same as C's). And so the chain maps $H_\bullet(C) \to C$ and $C \to H_\bullet(C)$ are quasi-isomorphisms. Furthermore, since the (co)kernel of $ds + sd$ is the homology complex, C is exact (acylic) iff $ds + sd$ is an isomorphism.

This is not true over an arbitrary ring R, so we give this a name.

1.4.1 Definition

A complex C is called **split** if there are maps $s_n\colon C_n \to C_{n+1}$ such that $d = dsd$. These maps s_n are called **splitting maps**. If C is further acylic, then C is said to be **split exact**.

1.4.2 Proposition

A chain complex C is split iff $C_n \cong Z_n \oplus B'_n$ with $Z_n = B_n \oplus H'_n$.

Proof

If C is split then define B'_n to be the image of $s_{n-1}d_n$. Since $dsd = d$, we have that $\ker sd = \ker d = Z_n$ and $sdsd = sd$, so sd is a projection operator whose image is B'_n and whose kernel is Z_n, thus $C_n = Z_n \oplus B'_n$ as desired. Similarly, ds is a projection operator whose image is B_n; restricting it to Z_n gives us $Z_n = B_n \oplus H'_n$.

If $C_n \cong Z_n \oplus B'_n$ and $Z_n = B_n \oplus H'_n$ then we have $B'_n \cong B_{n-1}$. And so we can map $C_n \to C_{n+1}$ along $B_n \cong B'_{n+1}$; $(x, y, z) \in B_n \oplus H'_n \oplus B'_n$ maps to $(0, 0, x) \in B_{n+1} \oplus H'_{n+1} \oplus B'_{n+1}$. Then dsd maps $(x, y, z) \in C_{n+1}$ to $(z, 0, 0) \in C_n$ to $(0, 0, z) \in C_{n+1}$ to $(z, 0, 0) \in C_n$, which is the same as $d(x, y, z)$; $dsd = d$. ■

1.4.3 Proposition

A chain complex C is split exact iff there exist maps such that $ds + sd$ is the identity.

Proof

If $ds + sd = \mathrm{id}$ then we get that $(ds + sd)d = d$, but the left-hand side is $dsd + sd^2 = dsd$, so C is split. And C is split then we showed that $C_n \cong B_n \oplus H'_n \oplus B'_n$ with ds, sd the projections onto B_n, B'_n respectively. C is then split exact iff $H_n(C) \cong H'_n = 0$, which is iff the projection onto $B_n \oplus B'_n$ is the identity. This is $ds + sd$, and so C is split exact iff $ds + sd$ is the identity, as desired. ■

1.4.4 Example

Consider the complex

$$\cdots \xrightarrow{\cdot 2} \mathbb{Z}/4\mathbb{Z} \xrightarrow{\cdot 2} \mathbb{Z}/4\mathbb{Z} \xrightarrow{\cdot 2} \cdots$$

This is an acylic complex: the image and kernel of multiplication by 2 are $\{0, 2\}$. But it is not split exact: $ds + sd$ can never be the identity.

Now suppose we have two chain complexes C, D and maps $s_n \colon C_n \to D_{n+1}$. Define $f_n \colon C_n \to D_n$ by $f_n = d_{n+1}s_n + s_{n-1}d_n$: $f = ds + sd$. Then notice that

$$df = d(ds + sd) = dsd = (ds + sd)d = fd$$

so $f = ds + sd$ is a chain map $C \to D$.

1.4.5 Definition

A chain map $f \colon C \to D$ is **null homotopic** if there are maps $s_n \colon C_n \to D_{n+1}$ such that $f = ds + sd$. The maps s_n are called a **chain contraction** of f.

Immediately we get:

1.4.6 Proposition

C is split exact iff the identity is null homotopic.

Since the sum of chain maps is a chain map, the sum of any chain map with a null homotopic map is again a chain map.

1.4.7 Definition

We say that two maps $f, g \colon C \to D$ are **chain homotopic** if their difference is null homotopic. If $f - g = sd + ds$ then the maps s_n are called a **chain homotopy** from f to g. Two maps $f \colon C \rightleftarrows D \colon g$ form a **chain homotopy equivalence** if their compositions are chain homotopic to the respective identities. C and D are **chain homotopy equivalent** if there exists a chain homotopy equivalence between them.

In particular f is null homotopic iff it is chain homotopic to the zero map.

1.4.8 Lemma

If $f\colon C \to D$ is null homotopic then the induced maps $f\colon H_n(C) \to H_n(D)$ are zero. And so if f, g are chain homotopic their induced maps are equal.

Proof

The latter claim follows from the first: the induced map $H_n(f-g)$ is equal to $H_n(f) - H_n(g)$. So if $f-g$ is null homotopic, then $H_n(f) - H_n(g) = 0$.

So suppose $f = ds + sd$, then for $a \in Z_n(C)$ we have $fa = dsa + sda = dsa \in B_n(D)$. Thus f maps cycles to boundaries, and so its induced map is zero. ■

1.4.9 Example

If C is split, then the chain complex $H_\bullet(C)$ and C are chain homotopy equivalent. Indeed, consider $f\colon C \to C$ given by $f = \mathrm{id} - ds - sd$ (the difference between a chain map and a null homotopy). Then $df = 0$, so f maps into Z_n, that is it defines a map $f\colon C \to H_\bullet(C)$ (alternatively recall that $C_n \cong B_n \oplus H_n \oplus B'_n$, this is the projection onto H_n). Then define $g\colon H_\bullet(C) \to C$ by inclusion (recalling that $C_n \cong B_n \oplus H_n \oplus B'_n$).

Then fg is just the identity on $H_\bullet(C)$. And gf is projection onto H_n; it is $\mathrm{id} - ds - sd$ which is homotopy equivalent to id. Thus f, g form a chain homotopy equivalence. ■

Notice that chain homotopy equivalence is an equivalence relation: it is clearly reflexive (taking $s = 0$) and symmetric (taking $-s$). And if $f - g = sd + ds$ and $g - h = s'd + ds'$ then $f - h = (s + s')d + d(s + s')$, so it is transitive. This equivalence relation preserves addition: if $f - f' = sd + ds$ and $g - g' = s'd + ds'$ then $(f + g) - (f' + g') = (s + s')d + d(s + s')$.

Let us construct the category K whose objects are chain complexes and whose morphisms $C \to D$ are equivalence classes of chain maps. We need to show that composition is well-defined: if $f - g = sd + ds$ and u is another chain map then $uf - ug = usd + uds = usd + dus$, so uf and ug are homotopic (similar for fu, gu). Thus if $f \sim g$ and $u \sim v$ we have $fu \sim gu \sim gv$, so composition is well-defined.

This turns K into an additive category; it has biproducts. It is not an Abelian category as it doesn't necessarily have kernels.

Since we showed that chain homotopic maps have the same image under homology functors, homology gives functors $\mathsf{K} \to \mathcal{A}$. In particular chain homotopic equivalent complexes have isomorphic homology (since homotopy equivalence is isomorphism in K):

1.4.10 Theorem

If C and D are chain homotopic equivalent, then $H_\bullet(C) \cong H_\bullet(D)$.

1.5 Mapping cones and cylinders

1.5.1 Definition

Let $f\colon B \to C$ be a map of chain complexes. The **mapping cone** of f is the chain complex $\mathrm{cone}(f)$ whose nth component is $B_{n-1} \oplus C_n$. The differential is given by $d(b, c) = (-d(b), d(c) - f(b))$.

For cochain complexes, the nth component is $B^{n+1} \oplus C^n$, and the differential has the same formula.

We can represent the differential by the matrix $\begin{pmatrix} -d_B & 0 \\ -f & +d_C \end{pmatrix}$.

1.5.2 Example

Let $\mathrm{cone}(C)$ denote the mapping cone of the identity map id_C; its differential is given by $d(b,c) = (-db, dc-b)$. Then $s(b,c) = (-c,0)$ is a splitting map $\mathrm{cone}(C)_n = C_{n-1} \oplus C_n \to C_n \oplus C_{n+1} = \mathrm{cone}(C)_{n+1}$:

$$(ds + sd)(b,c) = d(-c,0) + s(-db, dc-b) = (dc, c) + (b - dc, 0) = (b,c)$$

so $\mathrm{cone}(C)$ is split exact.

1.5.3 Proposition

Let $f\colon C \to D$ be a map of complexes. Then f is null homotopic iff it extends to a map $(-s,f)\colon \mathrm{cone}(C) \to D$.

Proof

Note that $(-s,f)$ is a chain map iff $d(-s,f)(b,c) = (-s,f)d(b,c)$ for all b,c. These are equal to

$$d(-s,f)(b,c) = -ds(b) + df(c), \qquad (-s,f)d(b,c) = (-s,f)(-db, dc-b) = sd(b) + fd(c) - f(b)$$

since f is a chain map $fd = df$, and so these are equal iff

$$-ds(b) = sd(b) - f(b)$$

i.e. iff $f = sd + ds$, as desired. ■

Consider a chain map $f\colon H_\bullet(B) \to H_\bullet(C)$, then there is a short exact sequence

$$0 \to C \xrightarrow{\iota} \mathrm{cone}(f) \xrightarrow{\delta} B[-1] \to 0$$

where the left map maps $c \in C_n$ to $(0,c) \in \mathrm{cone}(f)_n$, and the right maps (b,c) to $-b$. These are chain maps: $d(0,c) = (0,dc)$ and $d(b,c) = (-db, dc - fb) \mapsto -db$ (recall the differential of $B[-1]$ is $-d_B$ by the sign convention). The left map is an injection, the right a surjection, and the image of the left is the kernel of the second (the component C_n).

And so this gives us a long exact sequence (recalling that $H_{n+1}(B[-1]) = H_n(B)$)

$$\cdots \to H_{n+1}(\mathrm{cone} f) \xrightarrow{\delta} H_n(B) \xrightarrow{\partial} H_n(C) \to H_n(\mathrm{cone} f) \xrightarrow{\delta} H_{n-1}(B) \to \cdots$$

Let us understand the connecting maps ∂. Recall that $\partial = \iota^{-1} d_{\mathrm{cone}(f)} \delta^{-1}$, so for a cycle $b \in B_n$ we can lift it to $(-b,0) \in \mathrm{cone}(f)$. Then applying the differential gives $(db, fb) = (0, fb)$ which maps to fb. Thus we have $\partial([b]) = f([b])$ for all $[b] \in H_\bullet(B)$, so our connecting map is simply f.

We have proven the following.

1.5.4 Proposition

For any map $f\colon H_\bullet(B) \to H_\bullet(C)$ we have a long exact sequence

$$\cdots \to H_{n+1}(\mathrm{cone} f) \to H_n(B) \xrightarrow{f} H_n(C) \to H_n(\mathrm{cone} f) \to H_{n-1}(B) \xrightarrow{f} \cdots$$

1.5.5 Corollary

$f\colon B \to C$ is a quasi-isomorphism iff cone$f$ is acyclic.

Proof

If cone$f$ is acyclic the above long exact sequence gives us exact sequences $0 \to H_n(B) \xrightarrow{f} H_n(C) \to 0$, so the induced map is an isomorphism. And if f is a quasi-isomorphism, we must have that π and δ are zero on the homology modules, since $\ker\delta = \mathrm{im}\pi = 0$ we must have that $H_n(\mathrm{cone}f) = 0$, so cone$f$ is acyclic. ■

We have a similar construction:

1.5.6 Definition

Let $f\colon B \to C$ be a chain map, then define the **mapping cylinder** of f to be the complex $\mathrm{cyl}(f)$ whose nth component is $B_n \oplus B_{n-1} \oplus C_n$. The differential is

$$d(b, b', c) = (d(b) + b', -d(b'), d(c) - f(b'))$$

We can represent the differential as the matrix

$$d = \begin{pmatrix} d_B & \mathrm{id}_B & 0 \\ 0 & -d_B & 0 \\ 0 & -f & d_C \end{pmatrix}$$

We see that this is a complex:

$$d^2 = \begin{pmatrix} d_B^2 & d_B - d_B & 0 \\ 0 & d_B^2 & 0 \\ 0 & fd_B - d_C f & d_C^2 \end{pmatrix} = 0$$

1.5.7 Proposition

Let $\mathrm{cyl}(C)$ be the mapping cylinder of id_C. Then $f, g\colon C \to D$ are chain homotopic iff they extend to a map $(f, s, g)\colon \mathrm{cyl}(C) \to D$.

Proof

(f, s, g) is a chain map iff $d(f, s, g)(b, b', c) = (f, s, g)d(b, b', c)$ for all b, b', c. These are equal to

$$d(f, s, g)(b, b', c) = df(b) + ds(b') + dg(c), \qquad (f, s, g)d(b, b', c) = (f, s, g)(d(b) + b', -d(b'), d(c) - b') = fd(b) + f(b') - sd(b') + gd($$

These are equal iff $f(b') - sd(b') - g(b') = ds(b')$ i.e. iff $f - g = ds + sd$, as desired. ■

1.5.8 Lemma

The subcomplex whose elements are of the form $(0, 0, c)$ is isomorphic to C and the corresponding inclusion $\alpha\colon C \to \mathrm{cyl}(f)$ is a quasi-isomorphism.

Proof

Clearly $\alpha: c \mapsto (0,0,c)$ is an injection whose image is this subcomplex. We need to show that α is a chain map:

$$d\alpha(c) = d(0,0,c) = (0,0,dc) = \alpha(dc)$$

as desired.

Now, up to isomorphism, $\mathrm{cyl}(f)/\alpha(C)$ has components of the form $B_n \oplus B_{n-1}$ and the differential is given by $d(b,b') = (d(b)+b', -d(b'))$. This is clearly isomorphic to $\mathrm{cone}(-\mathrm{id}_B)$, which we showed is split exact (we showed $\mathrm{cone}(\mathrm{id}_B)$ is). So we have a short exact sequence

$$0 \to C \xrightarrow{\alpha} \mathrm{cyl}(f) \to \mathrm{cone}(-\mathrm{id}_B) \to 0$$

and the homology sequence gives us $0 = H_{n+1}(\mathrm{cone}(-\mathrm{id}_B)) \to H_n(C) \xrightarrow{\alpha} H_n(\mathrm{cyl}(f)) \to H_n(\mathrm{cone}(-\mathrm{id}_B)) = 0$, that is to say α is a quasi-isomorphism. ■

We can actually strengthen this:

1.5.9 Lemma

α is actually a chain homotopy equivalence (thus an isomorphism in K, so in particular a quasi-isomorphism).

Proof

Define $\beta: \mathrm{cyl}(f) \to C$ by $\beta(b,b',c) = f(b) + c$. This is a chain map:

$$d\beta(b,b',c) = df(b)+dc, \qquad \beta d(b,b',c) = \beta(d(b)+b', -d(b'), d(c)-f(b')) = f(db+b')+dc-f(b') = fd(b)+dc$$

Now $\beta\alpha = \mathrm{id}_C$ and defining $s_n: \mathrm{cyl}(f)_n \to \mathrm{cyl}(f)_{n+1}$ by $s(b,b',c) = (0,b,0)$ gives a chain homotopy from $\alpha\beta$ to id:

$$ds(b,b',c)+sd(b,b',c) = d(0,b,0)+s(d(b)+b', -d(b'), d(c)-f(b')) = (b, -d(b), -f(b))+(0, d(b)+b', 0) = (b, b', -f(b))$$

and

$$\alpha\beta(b,b',c) = (0,0,f(b)+c)$$

thus $\mathrm{id} - \alpha\beta = ds + sd$ as desired. ■

To summarize, $\alpha: C \rightleftarrows \mathrm{cyl}(C): \beta$ form a chain homotopy equivalence where

$$\alpha(c) = (0,0,c), \qquad \beta(b,b',c) = f(b) + c$$

Now consider the map $\iota: B \to \mathrm{cyl}(f)$ by $\iota(b) = (b,0,0)$. This is a chain map:

$$d\iota(b) = d(b,0,0) = (db,0,0) = \iota(db)$$

and so the subcomplex of $\mathrm{cyl}(f)$ whose elements are of the form $(b,0,0)$ is isomorphic to B. And the quotient complex $\mathrm{cyl}(f)/\iota(B)$ is isomorphic to $\mathrm{cone}(f)$: its terms are of the form $B_{n-1} \oplus C_n$ and the differential is given by $d(b',c) = (-d(b'), d(c) - f(b'))$.

Consider then the composite $B \xrightarrow{\iota} \mathrm{cyl}(f) \xrightarrow{\beta} C$: this is just f. Thus homology $f: H(B) \to H(C)$ factors through $H(B) \to H(\mathrm{cyl}(f))$. Then we can construct a commutative diagram:

$$\begin{tikzcd}
 & & C & & & \\
0 \arrow[r] & B \arrow[r] \arrow[ur, "f"] & \mathrm{cyl}(f) \arrow[u, "\beta"'] \arrow[r] & \mathrm{cone}(f) \arrow[r] \arrow[d, leftrightarrow, "="] & 0 & \\
0 \arrow[r] & & C \arrow[u, "\alpha"] \arrow[r] & \mathrm{cone}(f) \arrow[r, "\delta"] & B[-1] \arrow[r] & 0
\end{tikzcd}$$

The rows are exact (the map $\mathrm{cyl}(f) \to \mathrm{cone}(f)$ is given by the isomorphism $\mathrm{cyl}(f)/\iota(B) \cong \mathrm{cone}(f)$). Thus we have long exact sequences (recall that the connecting morphisms of the bottom homology are the induced maps of f):

$$\begin{tikzcd}
\cdots \arrow[r, "-\partial"] & H_n(B) \arrow[r] \arrow[d, leftrightarrow, "="] \arrow[dr, "f"] & H_n(\mathrm{cyl}(f)) \arrow[r] & H_n(\mathrm{cone}(f)) \arrow[r, "-\partial"] & H_{n-1}(B) \arrow[r] \arrow[d, leftrightarrow, "="] & \cdots \\
\cdots \arrow[r] & H_{n+1}(B[-1]) \arrow[r, "f"'] & H_n(C) \arrow[r] \arrow[u, leftrightarrow, "\cong"'] & H_n(\mathrm{cone}(f)) \arrow[r, "\delta"'] \arrow[u, leftrightarrow, "="] & H_n(B[-1]) \arrow[r, "f"'] & \cdots
\end{tikzcd}$$

We take the inverse of the top connecting map precisely because we want this diagram to commute.

To check that this diagram commutes we need only check that the right square with $-\partial$ and δ does. ∂ works as follows: let $[(b,c)]$ be a cycle in $\mathrm{cone}(f)$, so $db = 0$ and $dc = fb$. Then we lift it to $\mathrm{cyl}(f)$ to $(0,b,c)$ and then apply the differential $d(0,b,c) = (b, -db, dc - fb) = (b,0,0)$. Thus ∂ maps $[(b,c)] \in H(B)$ to $[b] \in H_{n-1}(B)$. δ maps $[(b,c)]$ to $[-b]$, i.e. $\delta = -\partial$ as desired.

And so these diagrams are isomorphic.

Now suppose we have a short exact sequence $0 \to B \xrightarrow{f} C \xrightarrow{g} D \to 0$. We define a chain map $\phi\colon \mathrm{cone}(f) \to D$ by $\phi(b,c) = g(c)$:

$$\phi d(b,c) = \phi(-db, dc - fb) = gdc - gfb = gdc = d\phi(b,c)$$

Then we have long exact sequences

$$\begin{tikzcd}
0 \arrow[r] & B \arrow[r, "f"] \arrow[d, leftrightarrow, "="] & C \arrow[r, "g"] & D \arrow[r] & 0 \\
0 \arrow[r] & B \arrow[r] & \mathrm{cyl}(f) \arrow[r] \arrow[u, "\beta"'] & \mathrm{cone}(f) \arrow[r] \arrow[u, "\phi"'] & 0 \\
 & 0 \arrow[r] & C \arrow[r] \arrow[u, "\alpha"'] & \mathrm{cone}(f) \arrow[r, "\delta"] \arrow[u, leftrightarrow, "="'] & B[-1] \arrow[r] & 0
\end{tikzcd}$$

Since our map from short to long exact sequences is a functor, and furthermore from the 5-lemma (a special case saying that if the rows of a two-row five-column commutative diagram are exact, and the outer four vertical maps are isomorphisms, then so is the innermost) ϕ is a quasi-isomorphism, since β is.

Thus the rows all generate isomorphic long exact sequences (though we must invert the connecting morphism of the top row, as per the previous argument).

1.5.10 Proposition

Let $f\colon B \to C$ be a chain map, and let $v\colon C \to \mathrm{cone}(f)$ be the inclusion $c \mapsto (0,c)$. There is a chain homotopy equivalence $\mathrm{cone}(v) \to B[-1]$.

Proof

The nth component of $\mathrm{cone}(v)$ is $C_{n-1} \oplus B_{n-1} \oplus C_n$ and its differential is

$$d(c',b,c) = (-d(c'), d(b,c) - v(c')) = (-d(c'), -d(b), d(c) - f(b) - c')$$

The maps $\alpha\colon \mathrm{cone}(v) \rightleftarrows B[-1]\colon \beta$ given by $\alpha(c',b,c) = b$ and $\beta(b) = (f(b),b,0)$ form a chain homotopy equivalence. $\alpha\beta = \mathrm{id}$ and $\beta\alpha(c',b,c) = (-f(b),b,0)$ is chain homotopic to id. Define $s_n\colon C_{n-1} \oplus B_{n-1} \oplus C_n \to C_n \oplus B_n \oplus C_{n+1}$ by $s_n(c',b,c) = (-c,0,0)$. Then

$$ds(c',b,c) + sd(c',b,c) = d(-c,0,0) + s(-dc', -db, dc - fb - c') = (dc,0,c) + (c' + fb - dc, 0, 0) = (c' + fb, 0, c)$$

which is equal to $(c',b,c) - \beta\alpha(c',b,c) = (c' + fb, 0, c)$, meaning id and $\beta\alpha$ are chain homotopic. ■

1.5.11 Example

Suppose C and C' are split complexes with splitting maps s, s'. Then $\sigma(c, c') = (-sc, s'c' - s'fsc)$ is a splitting map of $\operatorname{cone}(f)$ for $f\colon C \to C'$ iff the induced map $H(f)$ is zero. Indeed:

$$d\sigma d(c, c') = d\sigma(-dc, dc' - fc) = d(sdc, s'dc' - s'fc + s'fsdc) = (-dc, ds'dc' - ds'fc + ds'fsdc - fsdc)$$

Since s' is split, $ds'd = d$ so this is equal to $(-dc, dc' - (ds'f - ds'fsd + fsd)c)$, which is equal to d iff $ds'f - ds'fsd + fsd = f$. Now since C is split, we can embed $H(C)$ into it with projection $\mathrm{id} - ds - sd$, and so $H(f)$ is zero iff $f = fds + fsd$. So if $H(f) = 0$ then we have

$$ds'f - ds'fsd + fsd = ds'(f - fsd) + fsd = ds'fds + fsd = dfs + fsd = f$$

(since $fd = df$) as desired. Now if σ is split then for $z \in Z_n(C)$ we have that $fz = ds'fz \in B_n(C')$, so f is zero on the homology modules.

1.6 More on Abelian categories

We begin by mentioning an important theorem in the study of Abelian categories, without proof:

1.6.1 Theorem (Freyd-Mitchell)

If $\mathcal{A}$ is a small Abelian category, then there exists an exact fully faithful functor from $\mathcal{A}$ to Mod_R for some ring R.

What is an *exact* functor? We will define this soon.

1.6.2 Lemma

Let $\mathcal{C} \subseteq \mathcal{A}$ be a full subcategory of the Abelian category $\mathcal{A}$.

(1) $\mathcal{C}$ is additive iff $\mathcal{C}$ has $\mathcal{A}$'s zero objects and is closed under $\mathcal{A}$'s biproducts.

(2) $\mathcal{C}$ is itself Abelian and an exact subcategory of $\mathcal{A}$ iff $\mathcal{C}$ is additive and closed under $\ker, \operatorname{coker}$.

1.6.3 Example

(1) Mod_R is Abelian and the subcategory of finitely-generated R-modules forms an additive category. It is Abelian iff R is Noetherian.

(2) Ab is Abelian, and the subcategory of torsion-free Abelian groups is an additive subcategory. The subcategory of p-groups is Abelian, so is the subcategory of finite p-groups.

1.6.4 Proposition

Let $\mathcal{A}$ be an Abelian category and $\mathcal{C}$ an arbitrary category. The **functor category** $[\mathcal{C}, \mathcal{A}]$ is Abelian.

Proof

The zero object in $[\mathcal{C}, \mathcal{A}]$ is the constant functor to $0 \in \mathcal{A}$. For two functors $F, G: \mathcal{C} \to \mathcal{A}$ define $F \oplus G: \mathcal{C} \to \mathcal{A}$ to map $F \oplus G(c) = Fc \oplus Gc$ and $f: c \to d$ to the matrix $\mathrm{diag}(Ff, Gf)$. This is a biproduct. The kernel of $\alpha: F \Rightarrow G$ is $(\ker\alpha)(c) = \ker(\alpha(c))$, similar for cokernels. ■

1.6.5 Definition

For a topological space X, consider the partial order $\mathcal{U}$ of open subsets of X, considered as a category. A **presheaf** (with values in $\mathcal{A}$) is a contravariant functor $F: \mathcal{U}^{\mathsf{op}} \to \mathcal{A}$ such that $F\emptyset = 0$. The full subcategory of $[\mathcal{U}^{\mathsf{op}}, \mathcal{A}]$ of presheaves is denoted $\mathsf{Psh}_{\mathcal{A}}(X)$ (some authors do not require $F\emptyset = 0$, in which case the category of presheaves is just the functor category). This is an Abelian category.

An example of a presheaf with values in $\mathrm{Mod}_{\mathbb{R}}$ is the functor C^0 mapping an open set U to the set $C^0(U)$ of continuous functions $U \to \mathbb{R}$. For an inclusion $U \subseteq V$, the morphism $C^0(V) \to C^0(U)$ is given by restricting functions to U.

For a presheaf F, for $U \subseteq V$ we call the induced morphism $FV \to FU$ a **restriction**, and for $f \in FV$ its image under this morphism is also denoted $f|_U$.

1.6.6 Definition

A **sheaf** on a topological space X is a presheaf F which satisfies the following **sheaf axiom**: let $\{U_i\}$ be an open cover of U, an open subset of X. If $\{f_i \in F(U_i)\}$ are such that f_i and f_j agree on $U_i \cap U_j$ (i.e. their restrictions onto $U_i \cap U_j$ are equal), then there is a unique $f \in F(U)$ whose restriction to U_i is f_i.

Note by uniqueness if $f|_{U_i} = g|_{U_i}$ for all i then $f = g$. The sheaf axiom is equivalent to saying that if $f \in F(U)$ vanishes on every U_i, then $f = 0$. Thus we can write the sheaf axiom in element-free form. It is equivalent to saying that for any open U and open cover $\{U_i\}_i$ of U, the following sequence is exact:

$$0 \to F(U) \to \prod_i F(U_i) \to \prod_{i,j} F(U_i \cap U_j)$$

the first map maps $f \in F(U)$ to $(f|_{U_i})_i$, the second maps $(f_i)_i$ to $(f_i|_{U_i \cap U_j} - f_j|_{U_i \cap U_j})_{i,j}$.

1.6.7 Example

Let A be an Abelian group. For any open subset U of X, consider $A(U)$ the set of continuous maps from U to the discrete topological space on A. $A(V) \to A(U)$ given by restriction forms this into a sheaf.

The category of sheaves, $\mathsf{sh}(X)$, is an Abelian category contained in $\mathsf{Psh}(X)$; but it is not an Abelian subcategory: cokernels in the two categories differ.

For a topological space X, let $\mathcal{O}$ be the sheaf which maps U to $\mathcal{O}(U)$ the Abelian group of continuous maps $U \to \mathbb{C}$. Similarly define $\mathcal{O}^\times(U)$ to be the Abelian group of continuous maps $U \to \mathbb{C}^\times$. Then we have a short exact sequence of sheaves:

$$0 \to \mathbb{Z} \xrightarrow{2\pi i} \mathcal{O} \xrightarrow{\exp} \mathcal{O}^\times \to 0$$

This is not exact in $\mathsf{Psh}(X)$, though, when $X = \mathbb{C}^\times$.

1.6.8 Definition

An additive functor $F: \mathcal{A} \to \mathcal{B}$ between Abelian categories is **left exact** (resp. **right exact**) if for every short exact sequence $0 \to A \to B \to C \to 0$ in $\mathcal{A}$, $0 \to FA \to FB \to FC$ (resp. $FA \to FB \to FC \to 0$) is exact in $\mathcal{B}$. F is then **exact** if it is both left- and right-exact (meaning it preserves short exact sequences).

1.6.9 Example

The inclusion $\mathsf{sh}(X) \to \mathsf{Psh}(X)$ is left-exact (but not right-exact). There is an exact functor $\mathsf{Psh}(X) \to \mathsf{sh}(X)$ called **sheafification**.

1.6.10 Proposition

An additive functor $F\colon \mathcal{A} \to \mathcal{B}$ is left exact (resp. right exact) iff the exactness of $0 \to A \to B \to C$ (resp. $A \to B \to C \to 0$) implies the exactness of $0 \to FA \to FB \to FC$ (resp. $FA \to FB \to FC \to 0$).

Proof

We prove this in categories of modules, and use Freyd-Mitchell to justify the extension to Abelian categories. $0 \to A \to B \to C$ is exact iff $0 \to A \to B \to C' \to 0$ is exact where C' is the image of the final map in C. If F is left exact then $0 \to FA \to FB \to FC'$ is exact, now we claim $FC' \to FC$ is a monomorphism and thus $0 \to FA \to FB \to FC$ is exact as well. Indeed, $0 \to C' \to C \to C/C' \to 0$ is exact and thus since F is left exact $0 \to FC' \to FC$ is exact, meaning $FC' \to FC$ is a monomorphism.

Now suppose F preserves exactness of sequences of the form $0 \to A \to B \to C$, then in particular when we can append $\to 0$. ■

1.6.11 Proposition

Let $\mathcal{A}$ be an Abelian category, then $\hom_{\mathcal{A}}(M, \bullet)$ is a left exact functor $\mathcal{A} \to \mathrm{Ab}$ for every $M \in \mathcal{A}$.

Proof

Suppose we have $0 \to A \xrightarrow{f} B \xrightarrow{g} C \to 0$ in $\mathcal{A}$, we want to show that the following is exact

$$0 \to \hom(M, A) \xrightarrow{f_*} \hom(M, B) \xrightarrow{g_*} \hom(M, C)$$

where $f_*\alpha = f \circ \alpha$. Note that if $f_*\alpha = 0$ then $\alpha = 0$ since f is monic, thus f_* is monic. Furthermore since $g \circ f = 0$, $g_* f_*(\alpha) = g \circ f \circ \alpha = 0$ so $g_* f_* = 0$. Now if $\beta \in \hom(M, B)$ is in g_*'s kernel then $g \circ \beta = 0$ so $\beta(M) \subseteq \ker g = \mathrm{im} f$, thus β factors through f, meaning it is in f_*'s image as desired. ■

1.6.12 Corollary

$\hom_{\mathcal{A}}(\bullet, M)$ is left exact.

Proof

$\hom_{\mathcal{A}}(\bullet, M) = \hom_{\mathcal{A}^{\mathrm{op}}}(M, \bullet)$.

1.6.13 Definition

We define the **Yoneda embedding** $\mathsf{y}\colon \mathcal{A} \to [\mathcal{A}^{\mathrm{op}}, \mathrm{Ab}]$ for any additive category by mapping $A \mapsto h_A = \hom_{\mathcal{A}}(\bullet, A)$ and $f\colon A \to B$ to $f_*\alpha = f \circ \alpha$.

Considering the Yoneda embedding in classical category theory shows that this is a fully faithful embedding. When $\mathcal{A}$ is Abelian y is left exact since $\mathsf{y}(A) = h_A$ is.

1.6.14 Lemma (Yoneda)

The Yoneda embedding reflects exactness: $A \xrightarrow{\alpha} B \xrightarrow{\beta} C$ in $\mathcal{A}$ is exact if for every M the following sequence is exact

$$\hom(M, A) \xrightarrow{\alpha_*} \hom(M, B) \xrightarrow{\beta_*} \hom(M, C)$$

Proof

Let $M = A$ then we have $\beta\alpha = \beta_*\alpha_*\mathrm{id} = 0$. Taking $M = \ker\beta$ we see that the inclusion $\iota\colon \ker\beta \to B$ satisfies $\beta_*\iota = \beta\iota = 0$. Thus there is $\sigma \in \hom(M, A)$ such that $\iota = \alpha_*\sigma = \alpha\sigma$, so $\ker\beta = \mathrm{im}\iota \subseteq \mathrm{im}\alpha$ as desired. ■

2 Derived Functors

2.1 δ-functors

Recall that morphisms of exact sequences are collections of morphisms which commute with everything. That is to say, if we have two exact sequences $\{A_n, f_n\}_n$ and $\{B_n, g_n\}_n$ (i.e. $f_n\colon A_n \to A_{n+1}$ and $\mathrm{im} f_n = \ker f_{n+1}$), a morphism $A \to B$ is a collection $\{s_n\colon A_n \to B_n\}_n$ such that $sf = gs$ (i.e. view the exact sequences as chain complexes).

2.1.1 Definition

A **homological** (resp. **cohomological**) **δ-functor** from $\mathcal{A}$ to $\mathcal{B}$ (Abelian categories) is a collection of additive functors $T_n\colon \mathcal{A} \to \mathcal{B}$ (resp. $T^n\colon \mathcal{A} \to \mathcal{B}$) for $n \geq 0$, along with morphisms $\delta_n\colon T_n(C) \to T_{n-1}(A)$ (resp. $\delta^n\colon T^n(C) \to T^{n+1}(A)$) for every short exact sequence $0 \to A \to B \to C \to 0$ in $\mathcal{A}$.

A δ-functor must satisfy the following (set $T_n = T^n = 0$ for $n < 0$):

(1) for a short exact sequence $0 \to A \to B \to C \to 0$, we get a long exact sequence

$$\cdots \to T_{n+1}(C) \xrightarrow{\delta} T_n(A) \to T_n(B) \to T_n(C) \xrightarrow{\delta} T_{n-1}(A) \to \cdots$$

(ascending indices for cohomological δ-functors.) In particular T_0 is right exact (since $T_{-1} = 0$) and T^0 is left exact (T_n, T^n in general are not: we have an exact sequence $T_n(A) \to T_n(B) \to T_n(C)$ but we cannot append 0).

(2) For a morphism of short exact sequences, δ commutes with the induced morphisms. That is, if we have a morphism between short exact sequences $0 \to A' \to B' \to C' \to 0$ and $0 \to A \to B \to C \to 0$, we get commutative squares

$$\begin{tikzcd} T_n(C') \arrow[r, "\delta"] \arrow[d] & T_{n-1}(A') \arrow[d] \\ T_n(C) \arrow[r, "\delta"] & T_{n-1}(A) \end{tikzcd}$$

for every $n \geq 0$.

2.1.2 Proposition

Homology gives a homological δ-functor $H_\bullet\colon \mathsf{Ch}_{\geq 0}(\mathcal{A}) \to \mathcal{A}$; cohomology gives a cohomological δ-functor $H^\bullet\colon \mathsf{Ch}^{\geq 0}(\mathcal{A}) \to \mathcal{A}$.

Proof

The δ morphisms are the connecting maps ∂. The axioms are satisfied because the connecting maps are functorial. ■

2.1.3 Proposition

Let $\mathcal{S}(\mathcal{A})$ be the category of short exact sequences in $\mathcal{A}$. If T is a δ-functor then δ_n is a natural transformation from T_n to T_{n-1} (viewed as functors mapping $0 \to A \to B \to C \to 0$ to T_nC and $T_{n-1}A$ respectively).

Proof

This is immediate from the second axiom of δ-functors. ■

2.1.4 Example

Let $p \in \mathbb{Z}$ then define a homological δ-functor $\mathrm{Ab} \to \mathrm{Ab}$ by $T_0(A) = A/pA$ and $T_1(A) = {}_pA = \{a \in A \mid pa = 0\}$ ($T_n = 0$ for all other n). To get the δ morphisms, the snake lemma on the diagram

$$\begin{tikzcd}
0 \arrow[r] & A \arrow[r] \arrow[d, "p"] & B \arrow[r] \arrow[d, "p"] & C \arrow[r] \arrow[d, "p"] & 0 \\
0 \arrow[r] & A \arrow[r] & B \arrow[r] & C \arrow[r] & 0
\end{tikzcd}$$

Since $\ker p = {}_pA$ and $\operatorname{coker} p = A/pA$ we get an exact sequence

$$0 \to {}_pA \to {}_pB \to {}_pC \xrightarrow{\delta} A/pA \to B/pB \to C/pC \to 0$$

Which is precisely the first axiom. Functorality of the connecting morphism gives us the second axiom. We can also view this as a cohomological δ-functor by swapping T_0 and T_1.

2.1.5 Definition

A morphism $S \to T$ of δ-functors is a system of natural transformations $S_n \to T_n$ (or $S^n \to T^n$) which commute with δ.

A homological δ-functor T is **universal** if given any other homological δ-functor S and a natural transformation $f_0\colon S_0 \to T_0$, we can uniquely extend f_0 to a morphism $\{f_n\}_n : S \to T$. Dually a cohomological δ-functor T is **universal** if given any other cohomological δ-functor S and natural $f_0\colon T^0 \to S^0$ it can be uniquely extended to $T \to S$.

2.1.6 Example

We will show later that homology and cohomology are universal.

2.1.7 Proposition

If $F\colon \mathcal{A} \to \mathcal{B}$ is exact, then $T_0 = F$ and $T_n = 0$ for $n \neq 0$ is a universal δ-functor (both homological and cohomological).

Proof

A quick check of the axioms shows that F is indeed a δ-functor (with $\delta = 0$). Let S be another δ-functor (suppose homological) and if $f_0\colon S_0 \to F$ is natural then define $f_n\colon S_n \to T_n = 0$ to be zero. This is the only way to extend f_0, and it is a morphism. ■

2.2 Projective resolutions

2.2.1 Definition

An object P in an Abelian category $\mathcal{A}$ is **projective** if it satisfies the following universal lifting property: given an epimorphism $g\colon B \to C$ and a map $\gamma\colon P \to C$ there exists a (not necessarily unique) map $\beta\colon P \to B$ such that $\gamma = g \circ \beta$. In a diagram:

$$\begin{tikzcd} & P \arrow[dl, dashed, "\exists\beta"'] \arrow[d, "\gamma"] & \\ B \arrow[r] & C \arrow[r] & 0 \end{tikzcd}$$

We are mostly concerned with **projective modules** (projective objects in Mod_R). Clearly free modules are projective: take a basis $(v_i)_i$ and map v_i to a preimage of $\gamma(v_i)$.

2.2.2 Proposition

Direct summands of projective modules are projective.

Proof

If $P = M \oplus M'$ is projective then any map $\gamma\colon M \to C$ can be extended to a map $\gamma\colon P \to C$ and the rest follows. ■

2.2.3 Proposition

An R-module is projective iff it is a direct summand of a free R-module (that is, M is projective iff there exists M' such that $M \oplus M'$ is free).

Proof

Direct summands of a free module are by the previous proposition projective.

Conversely $F(A)$ be the free R-module on the set A, for a module M we can consider $F(M)$ on the set underlying M and we have a surjection $\pi\colon F(M) \to M$. If M is projective then the universal lifting property gives us a map $i\colon M \to F(M)$ such that $\pi i = \mathrm{id}_M$:

$$\begin{tikzcd} & M \arrow[dl, dashed, "\exists i"'] \arrow[d, "\mathrm{id}_M"] & \\ F(M) \arrow[r, "\pi"] & M \arrow[r] & 0 \end{tikzcd}$$

So $i\pi$ is a projection onto $i(M) \cong M$, meaning M is a direct summand of $F(M)$. ■

Over many nice rings (e.g. $\mathbb{Z}$, division rings), projective modules are all free. This is not always the case:

2.2.4 Example

If $R = R_1 \times R_2$ is the product of two rings then $R_1 \times 0, 0 \times R_2$ are projective as their sum is R (a free R-module). $P = R_1 \times 0$ is not free since $(0,1)P = 0$ (meaning for any choice $B \subseteq P$ there is an $r \in R$ such that $rB = 0$ so B cannot be linearly independent). So for example $\mathbb{Z}/6\mathbb{Z} \cong \mathbb{Z}/2\mathbb{Z} \times \mathbb{Z}/3\mathbb{Z}$ is projective but not free.

2.2.5 Example

Let k be a field and consider the matrix ring $R = M_n(k)$. This has a module k^n (multiplication is given by vector-matrix multiplication), and as an R-module we have $R \cong k^n \oplus \cdots \oplus k^n$, so k^n is projective. A free R-module needs to have dimension dn^2 over k for some $d > 0$, but $\dim_k k^n = n$ so it cannot be free.

The category of finite Abelian groups has no projective objects (other than 0): non-trivial subgroups of free Abelian groups are torsion-free and thus infinite.

2.2.6 Definition

$\mathcal{A}$ has **enough projectives** if for every $A \in \mathcal{A}$ there is an epimorphism $P \to A$ with P projective.

Mod_R has enough projectives, take the epimorphism $F(M) \to M$.

2.2.7 Lemma

M is projective iff $\hom_A(M, \bullet)$ is exact.

Proof

Suppose $\hom(M, \bullet)$ is exact, and we have a surjection $g\colon B \to C$ and a map $\gamma\colon M \to C$. Since g is surjective and $\hom(M, \bullet)$ is exact, g_* is surjective as well. Thus we can lift γ to β such that $\gamma = g_*\beta = g \circ \beta$; M has the universal lifting property, it is projective.

Recall that in general $\hom(M, \bullet)$ is left exact. So to show that it is exact, we must only show that for $0 \to A \to B \xrightarrow{g} C \to 0$, g_* is surjective as well (i.e. we can append $\to 0$). This is precisely what the universal lifting property says. ■

2.2.8 Definition

A chain complex where every component is projective is called a **chain complex of projectives**. It is not necessarily projective in Ch.

2.2.9 Proposition

A chain complex P is projective in Ch iff it is a split exact complex of projectives.

Proof

If P is projective then so is $P[-1]$: a surjection $B \to C$ and a map $P[-1] \to C$ can be shifted up to $B[1] \to C[1]$ and $P \to C[1]$, then the resulting lift can be shifted back down $P[-1] \to B$ ($[p]$ is functorial). We have a short exact sequence

$$0 \to P \to \operatorname{cone}(P) \to P[-1] \to 0$$

Now a surjection onto a projective object gives rise to a projection: we can lift $\mathrm{id}_{P[-1]}$ to $\beta\colon P[-1] \to \operatorname{cone}(P)$ such that $\delta\beta = \mathrm{id}_{P[-1]}$ where δ is the surjection. Then $\beta\delta$ is a projection onto $P[-1]$ whose kernel is P, thus $\operatorname{cone}(P) \cong P \oplus P[-1]$. Since $\operatorname{cone}(P)$ is exact, so too must P be.

Since $\delta(p,q) = -p$ we have that β is of the form $\beta(p) = (-p, s(p))$ for $s\colon P[-1] \to P$ (not necessarily a chain map). Now we claim that s is a splitting map. Since β is a chain map we have

$$d\beta(p) = d(-p, s(p)) = (dp, ds(p) + p), \qquad \beta(-dp) = (dp, -s(dp))$$

So we have $ds + \mathrm{id} = -sd$ so $\mathrm{id} = -sd - ds$ meaning P is split exact.

Now we must show that P's components are all projectives. So we need to show that each B'_n is projective: let $g\colon N \to M$ surjective and $\gamma\colon B'_n \to M$. Considering N and M as chain complexes centered at n with all other terms 0, g gives rise to a surjection of a chain complex, and γ a map from P to M. Then we can lift it to $\beta\colon P \to N$, and we get the desired result.

Recall that if P is split exact then $P_n \cong B_n \oplus B'_n$ for $B'_n \cong B_{n-1}$. The differential is zero on B_n and an isomorphism $B'_n \cong B_{n-1}$, so we can view $P_n \cong B_n \oplus B_{n-1}$ with $d(b,b') = (b',0)$. Now a map $\gamma\colon P \to C$ is of the form $(\gamma_n, \gamma'_n)\colon P_n \to C_n$ and this is a chain map iff $d\gamma'_n = \gamma_{n-1}$. Indeed: $(\gamma_{n-1}, \gamma'_{n-1})d(b,b') = \gamma_{n-1}(b')$ and $d(\gamma_n(b) + \gamma'_n(b')) = d\gamma_n b + d\gamma'_n b'$. This is true for all b, b' and so take $b = 0$ and we get $\gamma_{n-1} = d\gamma'_n$. And conversely this implies $d\gamma_n = 0$ and so we have equality.

So if P is split exact of projectives, let $(\gamma_n, \gamma'_n)\colon P \to C$ and $g\colon A \to C$ surjective. Each B_n is projective (as it is a direct summand of a projective), so we have maps $\beta_n\colon B_{n-1} \to A_n$ which lift γ'_n along g_n. Notice then that the map $d\beta_{n+1}\colon B_n \to A_n$ lifts γ_n along g_n:

$$g_n d\beta_{n+1} = dg_{n+1}\beta_{n+1} = d\gamma'_{n+1} = \gamma_n$$

Then the map $(d\beta_{n+1}, \beta_n)\colon P \to A$ is a chain map and it lifts $\gamma + \gamma'$ along g as desired. ■

2.2.10 Corollary

If $\mathcal{A}$ has enough injectives, so does $\mathsf{Ch}(\mathcal{A})$.

Proof

Let C be a chain complex, then for each C_n let $g_n\colon B_n \to C_n$ be a surjection from a projective module. Then define $P_n = B_n \oplus B_{n-1}$ with differential mapping $d(b,b') = (b',0)$. This is a split exact complex of projectives, thus a projective complex. Now $(dg_{n+1}, g_n)\colon P \to C$ is a chain map as discussed in the previous proof, and is surjective as desired. ■

2.2.11 Definition

A short exact sequence $0 \to A \to B \to C \to 0$ is called **split** if there is an isomorphism $f\colon B \to A \oplus C$ that commutes with the short exact sequence (with inclusion $\iota\colon A \to A \oplus C$ and projection $\pi\colon A \oplus C \to C$). Explicitly, the following diagram commutes:

$$\begin{tikzcd}
0 \arrow[r] & A \arrow[r] \arrow[dr, "\iota"'] & B \arrow[r] \arrow[d, "f"] & C \arrow[r] & 0 \\
& & A \oplus C \arrow[ur, "\pi"'] & &
\end{tikzcd}$$

2.2.12 Lemma (Splitting lemma)

For a split exact sequence $0 \to A \xrightarrow{f} B \xrightarrow{g} C \to 0$, the following are equivalent:

(1) the sequence splits;

(2) the sequence is **left split**: there exists a morphism $t\colon B \to A$ such that $tf = \mathrm{id}_A$;

(3) the sequence is **right split**: there exists a morphism $s\colon C \to B$ such that $gs = \mathrm{id}_C$.

Proof

The sequence being split implies being left and right split: define $t\colon A \oplus C \cong B \to A$ to project onto A and $s\colon C \to B \cong A \oplus C$ to embed C into B. If the sequence is left split then $B = \ker t \oplus \operatorname{im} f$ since their intersection is zero and $b = (b - ft(b)) + ft(b)$. Now $\operatorname{im} f \cong A$ and $\ker t \cong C$ since we can map $b \in \ker t$ to $g(b) \in C$: this map is injective since its kernel is $\ker t \cap \ker g = \ker t \cap \operatorname{im} f = 0$. It is surjective since $g(b - ft(b)) = g(b)$. Thus $B \cong A \oplus C$ and this commutes with the sequence. ■

Note that this lemma is not true in the category of groups: being (left) split is equivalent to B being the direct product of A and C, but being right split is equivalent to B being the semidirect product of A and C. Thus (left) split implies right split, but not the other way around.

2.2.13 Proposition

An object $P \in \mathcal{A}$ is projective iff every short exact sequence $0 \to A \to B \to P \to 0$ is split.

Proof

If P is projective then we can lift id_P along the surjection $B \to P$ to give us a map which composes with it to give the identity. Thus the sequence is right split, and thus split.

Conversely suppose every such short exact sequence is split. Now $\mathcal{A}$ is finitely complete and thus for a surjection $A \to B$ and a map $P \to B$ we can consider the diagram's pullback C:

$$\begin{tikzcd}
C \arrow[r, "f"'] \arrow[d] & P \arrow[r] \arrow[d] & 0 \\
A \arrow[r] & B \arrow[r] & 0
\end{tikzcd}$$

It can be shown that the first row is indeed exact (f is epi), since in Abelian categories epimorphisms are regular (cokernels of their kernels), and we can follow the kernel of the bottom map. Taking K to be the

kernel of f, we have an exact sequence $0 \to K \to C \xrightarrow{f} P \to 0$, which must split. Then there is a map $g\colon P \to C$ such that $fg = \mathrm{id}_P$. Now consider $P \xrightarrow{g} C \to A$: by commutativity we have that it gives the desired lifting of $P \to B$. ■

2.2.14 Definition

Let $M \in \mathcal{A}$, a **left resolution** of M is an exact sequence

$$\cdots \xrightarrow{d} P_1 \xrightarrow{d} P_0 \xrightarrow{\epsilon} M \to 0$$

The map ϵ is called the **augmentation map**. This is a **projective resolution** if each P_i are projective.

2.2.15 Lemma

If $\mathcal{A}$ has enough projectives, every object M has a projective resolution. In particular every R-module has a projective resolution.

Proof

First take a projective P_0 and a surjection $\epsilon_0\colon P_0 \to M$, and set $M_0 = \ker\epsilon_0$. Inductively given M_{n-1} choose a projective P_n and surjection $\epsilon_n\colon P_n \to M_{n-1}$. Define $M_n = \ker\epsilon_n$ and let d_n be the composite $P_n \to M_{n-1} \to P_{n-1}$. Since $M_{n-1} \to P_{n-1}$ is monic $\ker d_n = \ker\epsilon_n = M_n$. And we have $d_n(P_n) = M_{n-1} = \ker d_{n-1}$, thus the resulting sequence is exact. ■

2.2.16 Proposition

A left resolution $\cdots \to P_1 \to P_0 \to M \to 0$ is the same as a quasi-isomorphism $P \to M$ where P is the chain complex $\cdots \to P_1 \to P_0 \to 0$ and M is considered as a chain complex focused at 0.

Proof

A chain map $P \xrightarrow{\epsilon} M$ is given by a map $P_0 \xrightarrow{\epsilon} M$ such that $\epsilon d = 0$ (for $d\colon P_1 \to P_0$). This is a quasi-isomorphism if $\epsilon\colon H_0(P) = P_0/dP_1 \to M$ is an isomorphism, meaning $\ker\epsilon = \mathrm{im}d$, i.e. the sequence is exact. ■

2.2.17 Theorem (Comparison theorem)

Let $P \xrightarrow{\epsilon} M$ be a projective resolution of M and $f'\colon M \to N$ a map in $\mathcal{A}$. Then for every resolution $Q \xrightarrow{\eta} N$, there is a chain map $f\colon P \to Q$ lifting f' in the sense that $\eta f_0 = f' \circ \epsilon$:

$$\begin{tikzcd}
\cdots \arrow[r] & P_1 \arrow[r] \arrow[d] & P_0 \arrow[r] \arrow[d] & M \arrow[r] \arrow[d, "f'"'] & 0 \\
\cdots \arrow[r] & Q_1 \arrow[r] & Q_0 \arrow[r] & N \arrow[r] & 0
\end{tikzcd}$$

f is unique up to chain homotopy equivalence.

Alternatively we could say that there exists a unique (up to chain homotopy equivalence) chain map between the chain complexes $P \to M$ to $Q \to N$.

The proof will show that $P \to M$ being a projective resolution is stronger a condition than necessary: it is sufficient to be given a chain complex $P \to M \to 0$ where P_n are all projective.

Proof

We construct f_n inductively; let $f_{-1} = f'$ and $f_{-2} = 0$. Suppose f_i has been constructed for $i \leq n$, so that $f_{i-1}d = df_i$. To construct f_{n+1} we consider the n-cycles of P, Q: if $n = -1$ then set $Z_{-1}(P) = M$ and $Z_{-1}(Q) = N$. Since $f_{n-1}d = df_n$ we know that f_n induces a map $f'_n \colon Z_n(P) \to Z_n(Q)$, so we have two diagrams with exact rows:

$$\begin{tikzcd}
\cdots \arrow[r] & P_{n+1} \arrow[r] \arrow[d, dashed] & Z_n(P) \arrow[r] \arrow[d, "f'_n"] & 0 \\
\cdots \arrow[r] & Q_{n+1} \arrow[r] & Z_n(Q) \arrow[r] & 0
\end{tikzcd}
\qquad
\begin{tikzcd}
0 \arrow[r] & Z_n(P) \arrow[r] \arrow[d, "f'_n"] & P_n \arrow[r] \arrow[d, "f_n"] & P_{n-1} \arrow[d, "f_{n-1}"] \\
0 \arrow[r] & Z_n(Q) \arrow[r] & Q_n \arrow[r] & Q_{n-1}
\end{tikzcd}$$

Using the first diagram and P_{n+1} being projective allows us to lift $f'_n d$ to $f_{n+1} \colon P_{n+1} \to Q_{n+1}$ such that $df_{n+1} = f'_n d = f_n d$. And so $\{f_n\}$ is a chain map extending f' as desired.

To show uniquess, suppose $g \colon P \to Q$ is a chain map extending f' as well. Let $h = f - g$ and we will construct a contraction $\{s_n \colon P_n \to Q_{n+1}\}$ inductively. For negative n we must have $s_n = 0$. For $n = 0$ we have $\eta h_0 = (f' - f')\epsilon = 0$, and so h_0 maps P_0 to $\ker\eta = Z_0(Q) = dQ_1$. And so we can lift h_0 to $s_0 \colon P_0 \to Q_1$ such that $h_0 = ds_0 = ds_0 + s_{-1}d$.

Inductively suppose for $i < n$ we have maps s_i such that $ds_{n-1} = h_{n-1} - s_{n-2}d$ and then consider $h_n - s_{n-1}d \colon P_n \to Q_n$. Note

$$d(h_n - s_{n-1}d) = dh_n - ds_{n-1}d = dh_n - (h_{n-1} - s_{n-2}d)d = dh_n - h_{n-1}d = 0$$

Thus $h_n - s_{n-1}d$ maps into $Z_n(Q) = dQ_{n+1}$, and so we can lift it to a map $s_{n+1} \colon P_n \to Q_{n+1}$ such that $ds_{n+1} = h_n - s_{n-1}d$. And so we have $h = ds + sd$ as desired. ■

Note that we did not use $P \to M$'s exactness: only $Q \to M$'s.

2.2.18 Lemma (Horeshoe lemma)

Suppose we have the following diagram:

$$\begin{tikzcd}
 & & & 0 \arrow[d] & \\
\cdots \arrow[r] & P'_1 \arrow[r] & P'_0 \arrow[r, "\epsilon'"] & A' \arrow[r] \arrow[d, "i_A"] & 0 \\
 & & & A \arrow[d, "\pi_A"] & \\
\cdots \arrow[r] & P''_1 \arrow[r] & P''_0 \arrow[r, "\epsilon''"] & A'' \arrow[r] \arrow[d] & 0 \\
 & & & 0 &
\end{tikzcd}$$

such that the column is exact and the rows are projective resolutions. Set $P_n = P'_n \oplus P''_n$, then P_n form a projective resolution P of A, and the colimn lifts to an exact sequence of complexes $0 \to P' \xrightarrow{i} P \xrightarrow{\pi} P'' \to 0$, where $i_n \colon P'_n \to P_n$ and $\pi_n \colon P_n \to P''_n$ are the inclusion and projection maps.

Proof

Lift ϵ'' to a map $P_0'' \to A$; then the direct sum of this map with $i_A\epsilon' \colon P_0' \to A$ gives a map $\epsilon \colon P_0 \to A$. Then the following diagram commutes:

$$\begin{tikzcd}
 & 0 \arrow[d] & 0 \arrow[d] & 0 \arrow[d] & \\
0 \arrow[r] & \ker\epsilon' \arrow[r] \arrow[d] & P_0' \arrow[r, "\epsilon'"] \arrow[d] & A' \arrow[r] \arrow[d] & 0 \\
0 \arrow[r] & \ker\epsilon \arrow[r] \arrow[d] & P_0 \arrow[r, "\epsilon"] \arrow[d] & A \arrow[r] \arrow[d] & 0 \\
0 \arrow[r] & \ker\epsilon'' \arrow[r] \arrow[d] & P_0'' \arrow[r, "\epsilon''"] \arrow[d] & A'' \arrow[r] \arrow[d] & 0 \\
 & 0 & 0 & 0 &
\end{tikzcd}$$

The two right columns are exact, and all the top and bottom rows are as well. The snake lemma gives us that the left column is exact, and $\operatorname{coker}\epsilon = 0$ (since $0 = \operatorname{coker}\epsilon' \to \operatorname{coker}\epsilon \to \operatorname{coker}\epsilon'' = 0$ is exact). Thus ϵ is epic, as desired.

Now we have another horseshoe:

$$\begin{tikzcd}
 & & & 0 \arrow[d] & \\
\cdots \arrow[r] & P_2' \arrow[r] & P_1' \arrow[r, "d'"] & \ker\epsilon' \arrow[r] \arrow[d] & 0 \\
 & & & \ker\epsilon \arrow[d] & \\
\cdots \arrow[r] & P_2'' \arrow[r] & P_1'' \arrow[r, "d''"] & \ker\epsilon'' \arrow[r] \arrow[d] & 0 \\
 & & & 0 &
\end{tikzcd}$$

and we can induct. ■

2.3 Injective resolutions

We can duallize the notion of projective objects:

2.3.1 Definition

An object I in an Abelian category $\mathcal{A}$ is **injective** if it satisfies the following universal lifting property: given a monomorphism $f \colon A \to B$ and a map $\alpha \colon A \to I$, we can lift α to a map $\beta \colon B \to I$ such that $\alpha = \beta \circ f$.

$$\begin{tikzcd} 0 \arrow[r] & A \arrow[r, "f"] \arrow[d, "\alpha"'] & B \arrow[dl, dashed, "\beta"] \\ & I & \end{tikzcd}$$

$\mathcal{A}$ has enough injectives if for every object $A \in \mathcal{A}$ there is an injective $I \in \mathcal{A}$ and a monomorphism $A \to I$.

Note that if $\{I_\alpha\}$ are injectives, so is their product (if it exists).

2.3.2 Lemma

An injective object in $\mathcal{A}$ is precisely a projective object in $\mathcal{A}^{\mathsf{op}}$. $\mathcal{A}$ has enough injectives iff $\mathcal{A}^{\mathsf{op}}$ has enough projectives.

2.3.3 Corollary

I is injective iff $\hom_{\mathcal{A}}(\bullet, I)$ is exact.

Proof

$I \in \mathcal{A}$ is injective iff $I \in \mathcal{A}^{\mathsf{op}}$ is projective iff $\hom_{\mathcal{A}^{\mathsf{op}}}(I, \bullet) = \hom_{\mathcal{A}}(\bullet, I)$ is exact. ■

2.3.4 Proposition (Baer's criterion)

A right R-module E is injective iff for every right ideal J of R, every map $J \to E$ can extend to a map $R \to E$.

Proof

If E is injective then we lift $J \to E$ along the injection $J \to R$. Conversely suppose we have an R-module B and a submodule A along with a map $\alpha\colon A \to E$. Let $\mathcal{E}$ be the poset of all extensions of α: maps $\alpha'\colon A' \to E$ for $A \subseteq A' \subseteq B$, partially ordered by $\alpha' \leq \alpha''$ iff α'' extends α'. This clearly satisfies the criteria for Zorn's lemma, as such it has a maximal element, $\alpha'\colon A' \to E$. We claim that $A' = B$.

Suppose not; there is some $b \in B$ not in A'. Define then $J = \{r \in R \mid br \in A'\}$: a right ideal of R. Then the composition $J \xrightarrow{b} A' \xrightarrow{\alpha'} E$ extends to a map $f\colon R \to E$. Let $A'' = A' + bR$, and define $\alpha''\colon A'' \to E$ by $\alpha''(a + br) = \alpha'(a) + f(r)$. This is well-defined since for $br \in A'$ we know $\alpha'(br) = f(r)$ by definition. As such α'' is a proper extension of α', a contradiction. ■

Recall that a semisimple R-module is one for which every submodule has a complement: if $A \leq M$ is a submodule then there exists $A' \leq M$ such that $M = A \oplus A'$. A semisimple ring is one which is semisimple as a module over itself (every ideal has a complement). Clearly then modules over semisimple rings are injective: let J' be J's complement and extend $J \to E$ to $R = J \oplus J' \to E$.

2.3.5 Corollary

Let R be semisimple, then every R-module is injective.

2.3.6 Corollary

Let R be a PID. Then an R-module M is injective iff it is **divisible**: for every nonzero $r \in R$ and $a \in M$ there is a $b \in M$ such that $a = br$.

Proof

If M is injective then for $r \in R$ and $a \in M$ we define $f: R \to M$ by $f(1) = a$ and extend by linearity: $f(x) = xa$. And define $g: R \to R$ by $g(1) = r$ (so $g(x) = rx$). Since R is an integral domain g is injective, and thus we can lift f to $f': R \to M$ such that $f'g = f$. Then $f'g(1) = f'(r) = rf'(1) = a$ as desired.

If M is divisible then for every $x \in R$, a map $f: (x) \to M$ is determined by its value on 1: $f(rx) = rxf(1)$. Then we can simply extend it to $f: R \to M$ by $f(r) = rf(1)$. All ideals of R are of this form (principal) since R is a PID, so by Baer M is injective as desired. ■

In particular $\mathbb{Q}, \mathbb{Z}_{p^\infty} = \mathbb{Z}[1/p]/\mathbb{Z}, \mathbb{Q}/\mathbb{Z}$ are injective Abelian groups ($\mathbb{Z}[1/p]$ are all rationals of the form a/p^n).

2.3.7 Proposition

Ab has enough injectives.

Proof

For an Abelian group A, define $I(A)$ to be the product of $\mathbb{Q}/\mathbb{Z}$ indexed by $\hom_{\mathrm{Ab}}(A, \mathbb{Q}/\mathbb{Z})$. This is a product of injectives and is thus also injective. Define $e_A: A \to I(A)$ by $e_A(a) = (f(a))_{f: A \to \mathbb{Q}/\mathbb{Z}}$.

We claim that e_A is an injection, so we must show that $\ker e_A = 0$. To do this we show that for every $0 \neq a \in A$, there is some $f: A \to \mathbb{Q}/\mathbb{Z}$ such that $f(a) \neq 0$. Let $\langle a \rangle \leq A$ be the subgroup generated by a, then by injectivity of $\mathbb{Q}/\mathbb{Z}$ we can lift any map $f: \langle a \rangle \to \mathbb{Q}/\mathbb{Z}$ to a map $A \to \mathbb{Q}/\mathbb{Z}$ (following the inclusion $\langle a \rangle \subseteq A$). Now $\langle a \rangle \cong \mathbb{Z}/m\mathbb{Z}$ or $\mathbb{Z}$ depending on a's order; in the first case define $f(a) = 1/m$ and in the second define $f(a) = 1/2$ say. In both cases these are well-defined maps where $f(a) \neq 0$, as desired. ■

Note that since $A \to (\mathbb{Q}/\mathbb{Z})^{\hom(A, \mathbb{Q}/\mathbb{Z})}$ is injective, if $\hom(A, \mathbb{Q}/\mathbb{Z}) = 0$ the right-hand side is zero and so too is A. Thus $A = 0$ iff $\hom(A, \mathbb{Q}/\mathbb{Z}) = 0$.

2.3.8 Proposition

$I \in \mathcal{A}$ is injective iff every short exact sequence $0 \to I \to A \to B \to 0$ splits.

Proof

By the duality between injectives and projectives. ■

2.3.9 Definition

Let $M \in \mathcal{A}$, then a **right resolution** of M is a cochain complex I with $I^n = 0$ for $n < 0$ and a map $M \to I^0$ such that the augmented complex

$$0 \to M \to I^0 \xrightarrow{d} I^1 \xrightarrow{d} \cdots$$

is exact. This is the same as a quasi-isomorphism $M \to I$. This is a **injective resolution** if each I_n is injective.

A right resolution in $\mathcal{A}$ is the same as a left resolution in A^{op}; an injective resolution in $\mathcal{A}$ is a projective resolution in $\mathcal{A}^{\mathsf{op}}$. Thus by dualizing our previous results we have the following:

2.3.10 Lemma

If $\mathcal{A}$ has enough injectives, every object in $\mathcal{A}$ has an injective resolution.

2.3.11 Theorem (Comparison theorem)

Let $N \to I$ be an injective resolution of N, and $f'\colon M \to N$ a map in $\mathcal{A}$. Then for every resolution $M \to E$ there is a cochain map $f\colon E \to N$ lifting f', unique up to chain homotopy equivalence.

2.3.12 Proposition

$I \in \mathsf{Ch}(\mathcal{A})$ is injective iff it is a split exact chain complex of injectives. Furthermore if $\mathcal{A}$ has enough injectives, so does $\mathsf{Ch}(\mathcal{A})$.

Proof

This follows immediately from previous results on projectives and the isomorphism $\mathsf{Ch}(\mathcal{A})^{\mathsf{op}} \cong \mathsf{Ch}(\mathcal{A}^{\mathsf{op}})$. This can be seen as follows: $\mathsf{Ch}(\mathcal{A})$ can be viewed as the functor category $[\mathbb{Z}_0, \mathcal{A}]$ (of additive functors) where $\mathbb{Z}_0$ is the category whose objects are integers and $\hom(i,j) = \mathbb{Z}$ when $j = i-1$ and 0 otherwise. Now in general $[\mathcal{C}, \mathcal{D}]^{\mathsf{op}} \cong [\mathcal{C}^{\mathsf{op}}, \mathcal{D}^{\mathsf{op}}]$ by mapping $F\colon \mathcal{C} \to \mathcal{D}$ to its opposite functor $F^{\mathsf{op}}\colon \mathcal{C}^{\mathsf{op}} \to \mathcal{D}^{\mathsf{op}}$ (which acts precisely as F does). Note that natural transformations $F^{\mathsf{op}} \Rightarrow G^{\mathsf{op}}$ are precisely natural transformations $G \Rightarrow F$, so this is a well-defined functor and clearly an isomorphism.

Note that $\mathbb{Z}_0^{\mathsf{op}} \cong \mathbb{Z}_0$ by mapping $n \mapsto -n$, and so

$$\mathsf{Ch}(\mathcal{A})^{\mathsf{op}} \cong [\mathbb{Z}_0, \mathcal{A}]^{\mathsf{op}} \cong [\mathbb{Z}_0, \mathcal{A}^{\mathsf{op}}] \cong \mathsf{Ch}(\mathcal{A}^{\mathsf{op}})$$

as desired.

Explicitly, we map a chain complex $C_\bullet$ to its cochain complex $C^\bullet$ (recall $C^n = C_{-n}$), and a chain map to its dual ($f^n = f_{-n}$). ■

We now will show that every category of modules has enough injectives. Note that since the opposite of a category of modules is itself not a category of modules, this is not a trivial result.

If A is an Abelian group and B a left R-module, then $\hom_{\mathrm{Ab}}(B, A)$ can be given a right R-module structure by $fr\colon b \mapsto f(rb)$.

2.3.13 Lemma

For every right R-module M, we have a natural isomorphism

$$\tau\colon \hom_{\mathrm{Ab}}(M, A) \to \hom_{\mathrm{Mod}_R}(M, \hom_{\mathrm{Ab}}(R, A))$$

given by $(\tau f)(m)\colon r \mapsto f(mr)$.

Proof

We define τ's inverse μ as follows: for $g\colon M \to \hom_{\mathrm{Ab}}(R, A)$ define $(\mu g)(m) = g(m)(1)$. For $f \in \hom_{\mathrm{Ab}}(M, A)$ then $(\mu\tau f)(m) = (\tau f)(m)(1) = f(m)$ so $\mu\tau$ is the identity. And for $f \in \hom_{\mathrm{Mod}_R}(M, \hom_{\mathrm{Ab}}(R, A))$ then $(\tau\mu f)(m)\colon r \mapsto (\mu f)(mr) = f(mr)(1)$. Notice $f(m)(r) = (f(m)r)(1) = (fmr)(1)$ as desired, so $\tau\mu f = f$ as

desired. ■

Thus $\hom_{\mathrm{Ab}}(R, \bullet)\colon \mathrm{Ab} \to \mathrm{Mod}_R$ is **right-adjoint** to the forgetful functor $U\colon \mathrm{Mod}_R \to \mathrm{Ab}$:

2.3.14 Definition

Functors $L\colon \mathcal{A} \rightleftarrows \mathcal{B}\colon R$ form an **adjunction** if there is a natural isomorphism

$$\tau_{AB}\colon \hom_{\mathcal{B}}(L(A), B) \cong \hom_{\mathcal{A}}(A, R(B))$$

(i.e. a natural isomorphism of the bifunctors given by the compositions of the hom-functors and L and R.) L is the **left adjoint** and R the **right adjoint**. This is commonly written $L \dashv R$.

2.3.15 Proposition

Suppose we have an adjunction of additive functors $L\colon \mathcal{A} \rightleftarrows \mathcal{B}\colon R$, then

(1) if L is exact and $I \in \mathcal{B}$ injective then $R(I) \in \mathcal{A}$ is injective as well;

(2) dually if R is exact and $P \in \mathcal{A}$ projective then $L(P) \in \mathcal{B}$ is projective.

Proof

We must show that $\hom_{\mathcal{A}}(\bullet, R(I))$ is exact, but by assumption this is isomorphic to $\hom_{\mathcal{B}}(L\bullet, I)$. Since I is injective, $\hom_{\mathcal{B}}(\bullet, I)$ is exact, and the composition of exact functors is obviously exact. Duality follows because $L \dashv R$ iff $R^{\mathsf{op}} \dashv L^{\mathsf{op}}$. ■

We showed that $U \dashv \hom_{\mathrm{Ab}}(R, \bullet)$ and the forgetful functor U is clearly exact. Thus $\hom_{\mathrm{Ab}}(R, I)$ is an injective R-module for any injective Abelian group I.

2.3.16 Theorem

Mod_R has enough injectives.

Proof

For an R-module M, define $I(M)$ to be the product of $I_0 = \hom_{\mathrm{Ab}}(R, \mathbb{Q}/\mathbb{Z})$ with itself indexed by $\hom_R(M, I_0)$. As explained, I_0 is injective and thus so is $I(M)$. Now define $e_M\colon M \to I(M)$ by $e_M(m) = (f(m))_{f:M\to I_0}$. Let $\alpha\colon R \to \mathbb{Q}/\mathbb{Z}$ be non-zero (which exists by extending a non-zero map $\langle 1\rangle \to \mathbb{Q}/\mathbb{Z}$ where $\langle 1\rangle$ is the subgroup generated by 1, which exists as per our previous argument). Now for $m \in M$ define $f\colon mR \to I_0$ by $f(m) = \alpha$, and since I_0 is injective this extends to $M \to I_0$ where $f(m) \neq 0$. Thus e_M has trivial kernel and is thus an injection. ■

2.3.17 Example

Consider $\mathsf{sh}(X)$, the category of sheaves (of Abelian groups) over a topological space X. We claim that $\mathsf{sh}(X)$ has enough injectives.

For a sheaf $\mathcal{F}$ and a point $x \in X$, $\mathcal{F}$'s **stalk** at x is the colimit $\mathcal{F}_x = \operatorname{colim}\{\mathcal{F}(U) \mid x \in U\}$ (i.e. the limit of the restriction of $\mathcal{F}$ to the poset of open subsets containing x). This is functorial $\mathsf{sh}(X) \to \mathrm{Ab}$ for every $x \in X$ (since colim is). If A is an Abelian group and $x \in X$, we define the **skyscraper sheaf** x_*A at

$x \in X$ to be the presheaf $(x_*A)(U) = A$ iff $x \in U$ and 0 otherwise. x_* acts on inclusions by either maps to zero or the identity, that is if $U \subseteq V$ then $\bullet|_U : (x_*A)(V) \to (x_*A)(U)$ is the identity if $x \in U$ and the zero map otherwise.

The skyscraper sheaf is indeed a sheaf. If $\{U_i\}$ is an open cover of U, suppose $x \in U$; then a_i, a_j ($a_i \in (x_*A)(U_i)$) agreeing on their intersection means that $a_i = a_j \in A$ if $x \in U_i, U_j$. Let a be the common element of all open sets containing x, this is the unique extension. Otherwise if $x \notin U$ all $a_i = 0$, so $a = 0$.

Note that $\{U \mid x \in U\}$ is a directed set and as such $\mathcal{F}_x$ is a direct limit which we can explicitly write out. $\mathcal{F}_x$ has underlying set $\bigsqcup \mathcal{F}(U)$ quotiented by an equivalence, where $s_i \in \mathcal{F}(U_i), s_j \in \mathcal{F}(U_j)$ are equated iff they have a common restriction (on some neighborhood of x).

Now we claim that $\mathcal{F}_x \dashv (x_*A)$ (functorial in x) for every sheaf $\mathcal{F}$ and Abelian group A. That is, we need a natural isomorphism

$$\hom_{\mathrm{Ab}}(\mathcal{F}_x, A) \cong \hom_{\mathsf{sh}(X)}(\mathcal{F}, x_*A)$$

Let $\phi : \mathcal{F}_x \to A$ be an Abelian morphism, then define $\phi'_U : \mathcal{F}(U) \to x_*(A)$ by $\phi'_U(s) = \phi([s])$ (where $[s]$ is the equivalence class of s in $\mathcal{F}_x$), this defines a morphism of sheaves $\phi' : \mathcal{F} \to x_*A$. Indeed: if $x \in U \subseteq V$ then $\phi'_U(s|_U) = \phi([s]) = \phi'_V(s)$ (the restriction of $x_*A(V)$ to $x_*A(U)$ is the identity. If $x \notin U$ then all maps compose to zero.

Conversely given $\psi : \mathcal{F} \to x_*A$ defining $\hat{\psi} : \mathcal{F}_x \to A$ by $\hat{\psi}([s]) = \psi_U(s)$ where $s \in \mathcal{F}(U)$. This is well-defined: if $s \in \mathcal{F}(U)$ and $s' \in \mathcal{F}(V)$ have the same restriction to a neighborhood of x, W ($s|_W = s'|_W$) then because ψ is a sheaf map we have that $\psi_U(s)|_W = \psi_V(s')|_W$. But restriction in the skyscraper sheaf is the identity: $\psi_U(s) = \psi_V(s')$ as desired. The composition of $\hat{\bullet}$ and $\bullet'$ gives the identities clearly, so we have a natural isomorphism, as desired.

Note that $\mathcal{F} \mapsto \mathcal{F}_x$ is an exact functor: for a morphism of sheaves $\phi : \mathcal{F} \to \mathcal{G}$ the induced map $\phi_x : \mathcal{F}_x \to \mathcal{G}_x$ is given by $\phi_x([s]) = [\phi_U(s)]$ for $s \in \mathcal{F}(U)$. The kernel of ϕ can be given by the sheaf $\ker\phi(U) = \ker\phi_U$, with restriction inherited from $\mathcal{F}$ (since $\ker\phi(U) \subseteq \mathcal{F}(U)$). And its image is given by $\mathrm{im}\phi(U) = \mathrm{im}\phi_U$, with restriction inherited from $\mathcal{G}$. Then $\ker\phi_x = \{[s] \in \mathcal{F}_x \mid s|_U = 0 \text{ for some } x \in U\} = \bigcup_{x \in U}[\ker\phi_U]$, and $\mathrm{im}\phi_x = \bigcup_{x \in U}[\mathrm{im}\phi_U]$. So if $\mathrm{im}\phi = \ker\psi$ we see that $\mathrm{im}\phi_x = \ker\psi_x$ as desired.

Thus x_*A is right-adjoint to an exact functor; it preserves injectives. So for every $x \in X$ choose an injective Abelian group, then x_*A_x is injective in $\mathsf{sh}(X)$, and thus so is $\prod_{x \in X} x_*A_x$.

Now given a sheaf $\mathcal{F}$, for every $x \in X$ choose an injection $\mathcal{F}_x \to I_x$ with I_x injective. Composing $\mathcal{F} \to x_*\mathcal{F}_x$ ($\phi_U(s) = [s]$ if $x \in U$ and 0 otherwise) with $x_*\mathcal{F}_x \to x_*I_x$ gives a map $\mathcal{F} \to \prod_{x \in X} x_*I_x = \mathcal{I}$. This is an injection: for $0 \neq s \in \mathcal{F}(U)$, note that there must be some $x \in V$ such that $s|_W \neq 0$ for all $W \subseteq V$ (by the sheaf axiom), and so $\phi_V(s) \neq 0$. Thus its kernel is zero.

So we have shown $\mathsf{sh}(X)$ has enough injectives.

2.3.18 Example

Let I be a small category and $\mathcal{A}$ Abelian and complete with enough injectives, we will show the functor category $\mathcal{A}^I = [I, \mathcal{A}]$ has enough injectives.

Consider the k**th coordinate functor** for $k \in I$: map $A : I \to \mathcal{A}$ to $A(k) \in \mathcal{A}$. This is an exact functor.

Now for $k \in I$ and $A \in \mathcal{A}$ define $k_*A : I \to A$ mapping $i \in I$ to

$$k_*A(i) = \prod_{\hom_I(i,k)} A$$

and if $f : i \to j$ is a map in I, the map $k_*A(f) : k_*A(i) \to k_*A(j)$ is determined by $f^* : \hom(j,k) \to \hom(i,k)$. That is, the coordinate $k_*A(i) \to A$ of this map corresponding to $\phi \in \hom_I(i,k)$ is the projection of $k_*A(i)$ onto the factor corresponding to $f^*\phi = \phi f$. And for $f : A \to B \in \mathcal{A}$ then define $k_*f : k_*A \to k_*B$ which maps each component $A \in k_*A$ to the corresponding term $B \in k_*B$ via f. This makes k_* a functor $\mathcal{A} \to \mathcal{A}^I$.

Now, k_* is right-adjoint to the kth coordinate functor:

$$\hom_{\mathcal{A}}(A(k), B) \cong \hom_{\mathcal{A}^I}(A, k_*B), \qquad A \in \mathcal{A}^I, B \in \mathcal{A}$$

Map $\phi\colon A(k) \to B$ to $\phi'\colon A \Rightarrow k_*B$ where $\phi'_i\colon A(i) \to k_*B(i)$ is defined by mapping $A(i) \xrightarrow{A(f)} A(k) \xrightarrow{\phi} B$ for each component $f\colon i \to k$. And for $\psi\colon A \Rightarrow k_*B$ map to $\hat{\psi}\colon A(k) \to B$ by the composition $A(k) \xrightarrow{\psi_k} k_*B(k) \xrightarrow{\pi_{\mathrm{id}}} B$ where π_{id} is the projection to the component associated with $\mathrm{id} \in \hom(k,k)$.

Thus k_* preserves injectives. For $F \in \mathcal{A}^I$, for each $k \in I$ embed $F(k) \to A_k$ for A_k injective. Then $E = \prod_{k\in I} k_*A_k$ exists (functor categories over complete categories are complete) and is injective. Further, $\eta\colon F \Rightarrow E$ is given by $\eta_i\colon F(i) \to \prod_{k\in I} k_*A_k(i)$ whose projection $F(i) \to A_k$ for $\phi\colon i \to k$ is given by the composition $F(i) \xrightarrow{F\phi} F(k) \to A_k$. This is a monomorphism.

Thus when $\mathcal{A}$ is complete Abelian and has enough injectives, so does $\mathcal{A}^I$.

Conversely if $\mathcal{A}$ is cocomplete and has enough projectives, $\mathcal{A}^{\mathsf{op}}$ has enough injectives and thus so does $[I^{\mathsf{op}}, \mathcal{A}^{\mathsf{op}}] \cong (\mathcal{A}^I)^{\mathsf{op}}$. Thus $\mathcal{A}^I$ has enough projectives.